Supplemental Information for "Open Wave Height Logger: an open source pressure sensor data logger for wave measurement"

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TABLE OF CONTENTS

Table of Figures **3**

INTRODUCTION 5

ELECTRONICS BUILD GUIDE 6

Ordering Circuit Boards **6**

Circuit Board Assembly **7**
CPU Board Assembly Steps 9
CPU Board Assembly Continued 10
Power Board Assembly Steps 12
Power circuit board assembly continued 13
MS5803 Sensor Board Build 15

Programming Newly Assembled OWHL Boards **16**
Modifying boards.txt 16
Burning the Bootloader 17
Connecting the USB to Serial Adapter 19
Setting the Real Time Clock 20
Programming a Serial Number Onto the OWHL 21
Programming the main OWHL data collection program 22
Choosing a microSD Card 22
Monitoring Battery Voltage 22
Monitoring Current Draw With an Oscilloscope 23

Battery Holder Options **26**
Attaching the JST Connector to the Metal Battery Holder 26
Attaching the Wire Leads 26
Attaching the JST Pigtail Connector 27
Modifying the Plastic Battery Holder to Fit Into the Housing 28

OWHL HOUSING BUILD 29

Making the Bulkhead **29**
Simple Build Instructions for Making the Bulkhead 30
Tapping the Hole for the 1/8" NPT Fitting 32
Gluing the MS5803 Circuit Board to the Bulkhead 33
Modifying the Cleanout Plug 36
Making the Oil Reservoir 37

Housing Assembly **39**
Oil Reservoir Installation 41
Filling the Oil Reservoir 41

Calibration **42**

ANCHORING METHODS 43

Anchoring In Soft Sediments **43**

Anchoring in Hard Reef **44**

Anchor Block Build **44**
Deck Block Modification 46
PVC pipe modification 46
Assembly of the Anchor Block 46

POST-PROCESSING INFORMATION 48

MATERIALS LISTS 49

WAVE SENSOR COMPARISON TABLE 57

Table of Figures

Figure 1. OWHL housing showing overlay of internals. 5
Figure 2. Left: top surface of the bare CPU circuit board. Right: top surface of the circuit board with all parts soldered. 9
Figure 3. Left: bottom surface of a bare CPU circuit board. Right: bottom surface of the CPU board with all parts installed. 10
Figure 4. Left: The bottom surface of a bare power circuit board. Right: The bottom surface of a power circuit board with all components installed. 12
Figure 5. The top surface of a fully assembled Power circuit board. The pins of the 1x6 headers are left at full length after soldering. 13
Figure 6. Left: The upper surface of a bare MS5803 sensor circuit board. Center: Top surface of the MS5803 board with MS5803-14BA sensor and 1x6 header pins installed. Right: Bottom surface of the MS5803 board with capacitor C5 and 1x6 headers installed. 15
Figure 7. An example of an AVR Pocket Programmer that serves as an In-System Programmer, which is used to burn a bootloader program onto a newly assembled OWHL. 17
Figure 8. Left: connection diagram for the In-System Program header pins on the OWHL CPU board. Right: Close up photo of the ISP header on CPU board. Pin 1 ("MISO") is located in the upper left of the header as pictured. 17
Figure 9. Using an Arduino Uno as ISP to burn the bootloader on a newly assembled OWHL CPU board. The OWHL board will be powered by the 3V3 and ground pins from the Arduino Uno in this setup. 18
Figure 10. FTDI Basic adapter from Sparkfun.com, 3.3 volt model. A set of 6 pins (shown) is useful for attaching the FTDI Basic to the female FTDI header on the bottom of the CPU circuit board or Power circuit board. 19
Figure 11. The green box highlights the FTDI header, viewed from the bottom surface of the Power circuit board. The left-most header pin is labeled 'DTR' and the right-most pin is labeled 'GND' and 'black' 19
Figure 12. FTDI Basic adapter attached to the Power board of an assembled OWHL stack. The 6-pin terminal on the end of the FTDI Basic adapter has a pin labeled "DTR" on the left, and a pin labeled "GND" on the right, as pictured. These pins should be matched to the pins labeled "DTR" and "GND" respectively on the OWHL 1x6 female FTDI header on the bottom of the Power or CPU circuit boards. 20
Figure 13. Example battery voltage depletion trends for three OWHL test units in the laboratory. All three devices were powered with 3 x "D" cell alkaline batteries, with an initial combined voltage of ~4.8V. The left most (green) curve illustrates the effect of using a microSD card that is not power efficient, while the two curves on the right used modern name brand SanDisk 16 GB cards purchased in 2014. 23
Figure 14. Inserting a 1 ohm resistor into the negative wire lead of the battery pack. A) Resistor and negative wire. B) The black wire is cut, stripped, and the 1 ohm resistor is soldered in place. C) A oscilloscope lead and ground clip are attached to the two legs of the 1 ohm resistor, and the battery pack is plugged into an OWHL programmed with the main data collection firmware. 24
Figure 15. A) Voltage trace from an oscilloscope showing the rise in voltage measured across a 1 ohm resistor corresponding to the increased current draw during a microSD card write event. The red arrows mark the start and end of the write event, which lasts approximately 60-75 ms. The short duration of the peak for the microSD card write event illustrates that the card quickly returns to a low power sleep mode after the write event. B) Voltage trace from a low quality microSD card showing a longer period of high voltage (marked by the red arrows), corresponding to high current draw during and after a write event. Here the voltage trace stays high for approximately 300 ms before reentering a low power state. 25
Figure 16. Detail of the terminal loop end of metal battery holder 26
Figure 17. Modified plastic three D-cell battery holder 28
Figure 18. Modified 3 D-cell battery holder with JST connector pigtail soldered on. 28
Figure 19. Exploded view of the OWHL housing components. 29
Figure 20. Bulkhead with oil reservoir and electronics. 29
Figure 21. A) Clamp the PVC sheet securely in a drill press. B) Drill partway through the PVC sheet using a 2" (51mm) diameter hole saw with a ¼" center drill. C) A partially drilled PVC sheet. D) Turn the PVC sheet over, and center the hole saw using the center hole from the previous drilling step. E) Finish drilling through the PVC sheet, and remove the bulkhead, using a small flatblade screwdriver to pry it loose if it becomes stuck in the hole saw. 30
Figure 22. Left: Use a size R drill bit to partially drill the existing center hole in the bulkhead. Right: The drilled hole in the bulkhead for the barbed fitting. This hole will have threads cut into it using a tap. 31

Figure 23. Left: Align the tap in the hole, ensuring that the tap stands perpendicular to the bulkhead. Slowly turn the tap in clockwise a few turns at a time by hand, and back the tap out counterclockwise every few turns to clear out the cut material. Right: The tapped threads are visible center hole, next to the barbed fitting. 33
Figure 24. Roughen the interior surface of the bulkhead where the pressure sensor circuit board will be glued. 33
Figure 25. A) Diagram side view of the assembled union fitting, with the tip of the 1/8" hose barb sticking out. B) Cross-section view of the union fitting. The OWHL pressure sensor board is glued to the bottom of the ¾" PVC bulkhead fitting with 3M 5200 adhesive, and the bulkhead fitting is glued into the upper half of the union fitting. 34
Figure 26. Left: apply adhesive to the surface of the bulkhead fitting on the side with the smaller hole. Right: apply adhesive to the top surface of the MS5803 circuit board, taking care to avoid getting adhesive on the white membrane of the sensor itself. 34
Figure 27. Left: roughened PVC bulkhead ready to receive adhesive. Center: MS5803 circuit board adhered to the bulkhead. Right: A view into the larger hole of the PVC bulkhead after mounting the MS5803 circuit board. The white membrane of the MS5803-14BA pressure sensor is visible in the center, and a layer of white adhesive has filled the space between the stainless steel ring of the pressure sensor and the PVC bulkhead hole. 35
Figure 28. A pair of c-clamps maintain light pressure on the MS5803 circuit board and the PVC bulkhead while the adhesive cures. 35
Figure 29. Views of the modified cleanout plug. The image on the right shows the copper mesh screen epoxied into place inside the plug. 36
Figure 30. Saline IV bag used to produce the oil reservoir. 37
Figure 31. Left: Using a heat sealer to seal the cut edge of an IV bag. Right: A modified IV bag, after re-sealing the trimmed edges of the cut-down bag. 38
Figure 32. Assembled OWHL housing. 39
Figure 33. An oil reservoir bag attached to the barbed fitting that mounts in the bulkhead. 42
Figure 34. Calculated water depth values reported by an OWHL versus known water depths in a calibration apparatus. The regression equation and R^2 for the best fit line between the expected and reported depth values is shown. 42
Figure 35. An OWHL attached to an 11 kg (25 lb) lead brick with band clamps. (Photo credit: The Bay Foundation) 43
Figure 36. Using sand anchors to secure sensors in soft sediments. (Photo credit, left - Aaron Ninokawa, right – The Bay Foundation) 43
Figure 37. Left: Stainless steel eye bolts epoxied into holes drilled in a rocky reef. Right: OWHL secured to an eye bolt using a stainless steel band clamp. (Photo credit: The Bay Foundation). 44
Figure 38. Left: Anchor block showing PVC pipe and SS lifting eye-bolt. Right: Anchor block with lifting line and subsurface buoy line attached. 45
Figure 39. An OWHL mounted inside 4 inch diameter pipe on a concrete anchor block. 45
Figure 40. The modified upper surface of the anchor block. 46
Figure 41. Inserting the eye bolts into the lag screw shield in the anchor block. 47

Introduction

The original design criteria and build goals for the Open Wave Height Logger focused on simplicity, durability, ease of construction and ease of use. In general, the OWHL is designed to be built from readily available materials and constructed using only basic shop tools found in most schools and field stations. Some of the build steps may require access to, or the purchase of, some specialized tools and equipment. In practice, we found that with some steps, more uniform results were achieved when better tools and equipment were available above what might be found in a simple workshop environment.

The OWHL housing is built using readily available, standard schedule 40, 1.5" PVC pipe and fittings (**Figure 1**). The custom OWHL electronics boards are designed to fit within this pipe (internal diameter of ~ 3.9 cm). The 1.5" pipe and the required pipe fittings can be found at nearly any home improvement, plumbing supply store, or like many parts of the OWHL, can be ordered online. Our design incorporates a break-apart union joint providing easy access to the internal components; allowing for simple battery changes and retrieval of the SD Card's data.

This guide starts with the assembly of the electronic components, followed by programming of the chips. The next steps involve the machining of the bulkhead fitting. This is followed by the modifications for the remaining housing parts and the assembly of the OWHL into its finished form. The guide finishes with a discussion of anchoring methods to secure the OWHL to the seafloor.

A materials list and a cut list are at the end of the instructions. Some dimensions in this document, particularly thread sizes and nominal pipe sizes, are given in Imperial units where there is not a directly interchangeable metric equivalent.

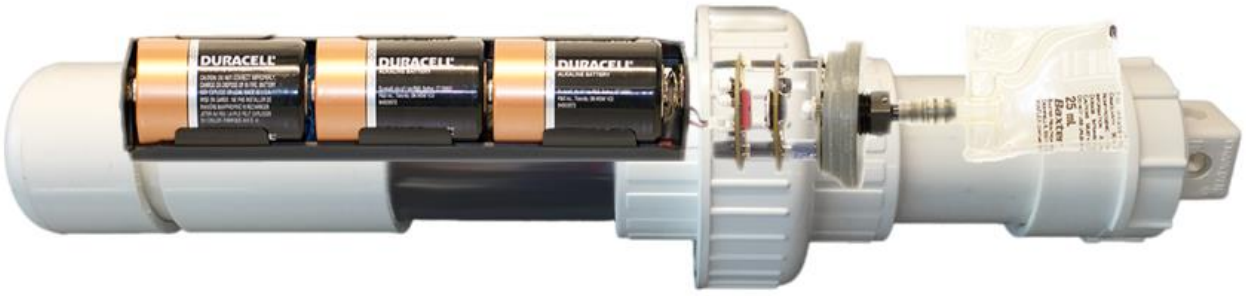


Figure 1. OWHL housing showing overlay of internals.

Electronics Build Guide

In the sections below we list required and optional tools and assembly materials, including multiple options where appropriate. Assembly of all three circuit boards proceeds by first attaching all of the surface-mount chips (resistors, capacitors, microcontroller etc.). When assembling the boards individually, hand soldering works well either with a temperature controlled soldering iron or hot air work station. When assembling the boards in large batches, we used an Arduino controlled reflow oven from plans found on the internet (link provided below). Once the surface mount chips have been attached, the various through-hole components (headers) and additional surface mount components found on the back sides of some boards can be hand soldered on to complete each board. We used this step to also preform quality control on the soldering by using a magnifying glass to visually check for bridged contacts. A video demonstration of the assembly steps is linked from the OWHL repository, https://github.com/millerlp/OWHL, and https://owhl.org (direct link: https://youtu.be/y3US9D-5TtE). A high quality version of the video is archived here: http://doi.org/10.5281/zenodo.3679029

Ordering Circuit Boards

The printed circuit boards can be ordered from a variety of online vendors by uploading the provided design files (industry standard 'gerber' files in a .zip archive). The files are found in the OWHL repository https://doi.org/10.5281/zenodo.3477584 under the Hardware_files/OWHL_revC/ directory. A separate .zip archive is provided for each of the three circuit boards:

- OWHL_CPU_disc_RevC_gerbers.zip
- OWHL_MS5803_disc_RevC_gerbers.zip
- OWHL_POWER_disc_RevC_gerbers.zip

Some recommended vendors for ordering circuit boards from include:

- https://oshpark.com
- https://www.seeedstudio.com/
- https://pcbshopper.com can provide price estimates from a wider variety of vendors

We have ordered solder paste stencils from http://oshstencils.com, which can also directly use the gerber files listed above to generate the stencil design for you.

Circuit Board Assembly

Tools Needed:

- Soldering Iron (temperature adjustable solder workstation preferred)
- Soldering Hot Air Re-work Station *(optional)*
- SMD Reflow oven *(optional)*
- Vise (to hold PCBs)
- Tweezers (to apply SMD parts to PCBs)

Materials needed:

- PCB Circuit Boards
- Solder Paste
- Rosin Core Solder (Lead-free preferred)
- PCB Circuit Board Stencils *(optional)*
- Plastic Scraper Card *(optional)*

For each PCB circuit board

- Carefully apply a tiny dot of solder paste to each pad that is labeled below
  OR
- Order a solder paste stencil when ordering the PCB circuit boards
    - Using the solder paste stencil
        - Position & align solder paste stencil carefully over the board pad locations
        - Apply solder paste to stencil
            - Drag a flat plastic card across the stencil holes to fill with solder paste
                - A credit card works well for this
            - Clear excess paste from the stencil with the card

The boards need to have the solder paste reach a temperature of 235 deg C to allow the solder to flow across the pads, cool down, solidify and anchor the parts to the circuit boards. This can be done in several ways:

- We used a reflow oven to complete the soldering process as it allows for making multiple boards at a time *(preferred method)*
    - Reflow ovens can be commercially purchased for ~$200 - $2000 (depending on size and wattage)
        - https://www.amazon.com/s?k=SMD+reflow+oven
    - An old toaster oven can be retrofitted with an Arduino controller to make an inexpensive reflow oven *(what we built and used)*
        - https://www.sparkfun.com/tutorials/60
- A hot air rework station can be used to hand solder the parts on boards
    - https://www.amazon.com/s?k=hot+air+rework+station
- A soldering iron workstation will be needed to complete the boards
    - Electronically controlled temperature versions of these are highly recommended
        - https://www.amazon.com/s?k=soldering+iron

Table 1. CPU circuit board parts list

CPU PCB Schematic Label	Part Description	Manufacturer	Mfr. Part Number	Quantity	digikey.com PN	newark.com PN	mouser.com PN
R9, R10	Resistor, surface-mount, 1k ohm, 1/8watt, 5% for SMD Piezo buzzer	Panasonic	ERJ-6GEYJ102V	2	P1.0KACT-ND	65T8778	667-ERJ-6GEYJ102V
R1, R2, R3, R4	Resistor, surface-mount 0805, 10k ohm, 1/8 watt 5%	Panasonic	ERJ-6GEYJ511V	4	P10KACT-ND	67T1248	667-ERJ-6GEYJ103V
CT10_REED_SWITCH	magnetic reed switch, surface mount	Coto	CT10-1030-G1	1	306-1258-1-ND	98H3468	816-CT10-1030-G1
U$3	Transistor (BJT) - NPN, for piezo buzzer, SOT-23-3 package	Fairchild	KSC1623LMTF	1	KSC1623LMTFCT-ND	99AC9286	512-KSC1623LMTF
PIEZO	3V piezo buzzer - SMD	Murata	PKLCS1212E4001-R1	1	490-4683-1-ND	01M8494	81-PKLCS1212E4001
J1	6-pin ISP header (3x2 layout)	3M	961206-6404-AR	1	3M9459-ND	89H8939	517-9612066404AR
U1	AVR ATMEGA328P-AU 32 PIN TQFP surface mount	AVR	ATMEGA328P-AU	1	ATMEGA328P-AU-ND	68T2935	556-ATMEGA328P-AU
IC1	DS3231SN Real-Time Clock 16SOIC-Wide	Maxim	DS3231SN	1	DS3231SN#T&RCT-ND	81Y9084	700-DS3231SN#
U$2	CR1220 Battery holder, surface-mount	Keystone	3000	1	36-3000-ND	66F5622	534-3000
C1, C2, C3	Capacitor, surface mount 0805 0.1uF 10V ceramic	Taiyo Yuden	LMK212SD104KG-T	4	587-1134-1-ND	84R9733	963-LMK212SD104KG-T
SPI_HDR, I2C_HDR, FTDI_HDR	Female wire-wrap headers (long-tail stacking headers for CPU + POWER discs) 10mm tail length	Samtec	SSQ-106-03-G-S	3	SAM1198-06-ND	11P8269	200-SSQ10603GS

CPU Board Assembly Steps

1. Use a solder paste stencil, or solder paste syringe, to apply solder paste to the solder pads on the top surface of the circuit board. Do not place solder on the through-hole pads, as these will be soldered separately.
2. Place the ATmega328p (CPU PCB parts list U1). The top plastic surface of the chip has a circular depression in one corner, which marks the location of pin 1. Align this circle with the white circle on the circuit board.
3. Place resistors 10k ohm resistors (CPU PCB parts list R1, R2, R3, R4) on the marked footprints R1, R2, R3, R4 (orientation does not matter).
4. Place 1k ohm resistors (CPU PCB parts list R9, R10) on the marked footprints R9, R10 (orientation does not matter).
5. Place the DS3231 real-time clock chip (CPU PCB parts list IC1). The plastic top of the chip has a semi-circular notch at one end, marking where pins 1 and 16 are. The circuit board footprint has a white dot next to pin 1. Place the chip so that pin 1 (top left of the chip looking down from the top) aligns with the pad marked by the white dot on the board.
6. Place the BJT transistor (CPU PCB parts list U$3) on its footprint between resistors R9 and R10.
7. Place 0.1 µF capacitors (CPU PCB parts list C1, C2, C3) on footprints C1, C2, C3 (orientation does not matter).
8. Place the reed switch (CPU PCB parts list CT10_REED_SWITCH). Orientation does not matter.
9. Solder the surface mount components using a reflow solder oven or hot air reflow station.

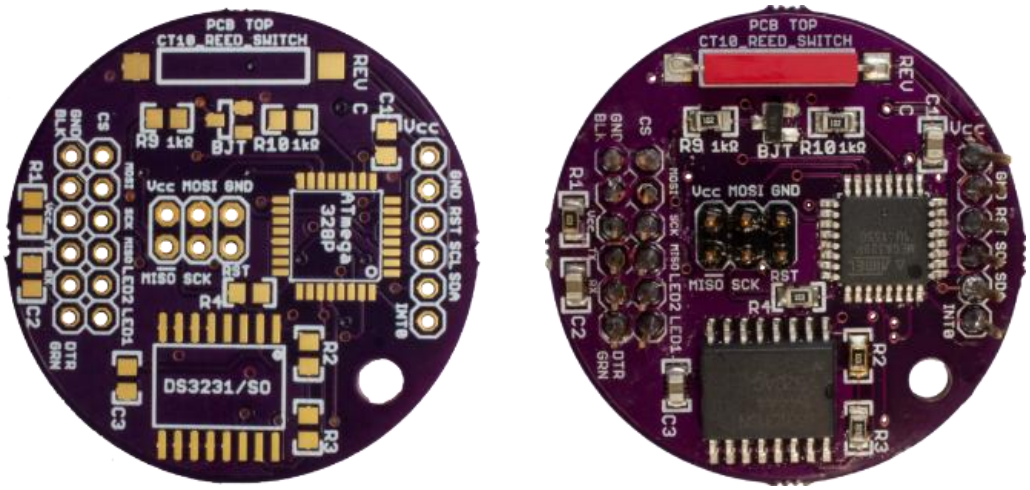


Figure 2. Left: top surface of the bare CPU circuit board. Right: top surface of the circuit board with all parts soldered.

CPU Board Assembly Continued

10. After the board has cooled, turn the board upside down (Figure 3). On the bottom side of the board, place a small layer of solder using a soldering iron on the pad in the middle of the battery holder footprint labeled CR1220.
11. Place the CR1220 battery holder (CPU PCB parts list U$2) on the footprint on the bottom surface of the CPU board. Make sure the open side of the holder faces the outer edge of the circuit board, following the outline pattern on the circuit board. Solder the tabs on each side of the battery holder in place using a soldering iron.
12. Place the piezo buzzer (CPU PCB parts list PIEZO) on the bottom surface of the circuit board. The solder pads on the buzzer match up with the footprint on the board in two orientations, either orientation can be used. Solder in place using a soldering iron.
13. Insert the 2x3 header (CPU PCB parts list J1) from the top side of the CPU circuit board. Solder the pins on the bottom side of the board using a soldering iron.
14. Insert 1x6 female wire-wrap headers (CPU PCB parts list SPI_HDR, I2C_HDR, FTDI_HDR) from the bottom side of the CPU board. Solder the pins on the top side of the board using a soldering iron. Do not clip the long pins of the headers.
15. Install a CR1220 battery into the holder, with the positive surface oriented upward, matching the + sign marked on the battery holder.

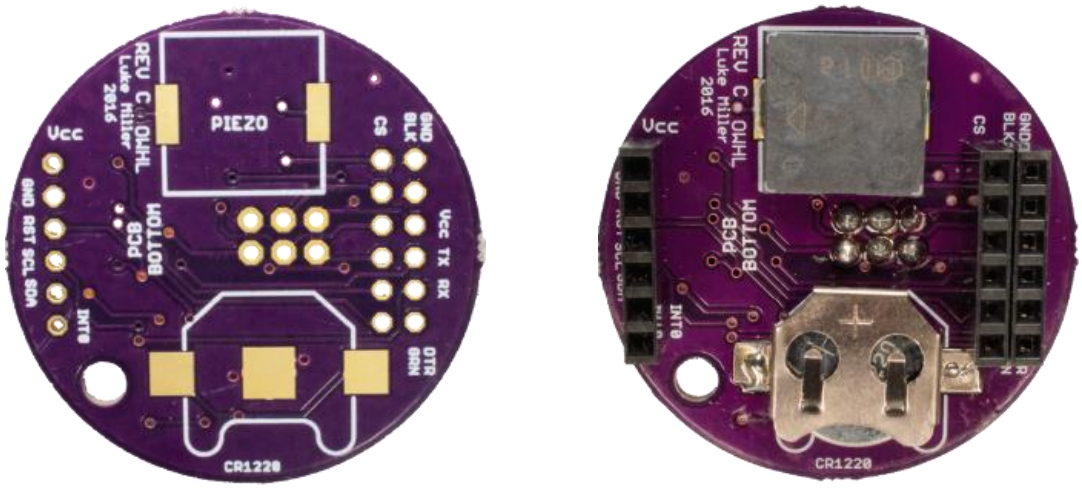


Figure 3. Left: bottom surface of a bare CPU circuit board.
Right: bottom surface of the CPU board with all parts installed.

Table 2. Power circuit board parts list.

Power PCB Schematic Label	Part Description	Manufacturer	Mfr. Part Number	Quantity	digikey.com PN	newark.com PN	mouser.com PN
LED1	RED LED, surface mount 0805, 1.8V forward voltage	Lite-On	LTST-C171KRKT	1	160-1427-1-ND	75R0581	859-LTST-C171KRKT
LED2	GREEN LED, surface mount 0805, 2.1V	Lite-On	LTST-C171GKT	1	160-1423-1-ND	75R0587	859-LTST-C171GKT
C3, C4	Capacitor, surface mount 1uF 35V, electrolytic, Panasonic FC series, size code B	Panasonic	EEE-FC1V1R0R	2	PCE4024CT-ND	53W3651	667-EEE-FC1V1R0R
C1	Capacitor, 22µF Tanatalum polarized 0805	AVX Corp	F920J226MPA	1	478-8119-1-ND	55AC2222	647-F920J226MPA
R7,R8	Resistor, surface-mount, 510 ohm, 1/8watt, 5%	Panasonic	ERJ-6GEYJ511V	2	P510ACT-ND	65T8831	667-ERJ-6GEYJ511V
R1,R2	Resistor, surface-mount, 100k ohm 1/8watt 5% for SD card pull ups	Panasonic	ERJ-6GEYJ104V	2	P100KACT-ND	65T8779	667-ERJ-6GEYJ104V
POWER_PLUG	JST PH series vertical 2-pin male header, SMD, 2mm spacing	JST	B2B-PH-SM4-TB	1	455-1734-6-ND	37K9941	
VREG	MCP1700T-3002E/TT 3.0V Regulator Surface-mount SOT-23-3	Microchip	MCP1700T-3002E/TT	1	MCP1700T-3002E/TTCT-ND	84R5176	579-MCP1700T3002E/TT
RESET, BUTTON1	pushbutton, surface-mount	Omron	B3S-1000P	2	SW836CT-ND	52F3560	653-B3S-1000P
U$2	MicroSD card slot, reverse mount	Molex	5027740891	1	WM24066CT-ND	71T7280	538-502774-0891
SPI_HDR, I2C_HDR, FTDI_HDR	Female wire-wrap headers (long-tail stacking headers for CPU + POWER discs) 10mm tail length	Samtec	SSQ-106-03-G-S	3	SAM1198-06-ND	11P8269	200-SSQ10603GS
	JST PH series 2-pin female connector, 2mm spacing	JST	PHR-2	1	455-1165-ND	28C6963	
	JST PH series crimp-on terminal for female connector housing, 24-30 AWG	JST	SPH-002T-P0.5S	6	455-1127-1-ND	28C6968	

Power Board Assembly Steps

1. Use a solder paste stencil, or solder paste syringe, to apply solder paste to the solder pads on the bottom surface of the Power circuit board. Do not place solder on the through-hole pads, as these will be soldered separately.
2. Place 1 µF capacitors (Power PCB parts list C3, C4) on positions C3, C4. Refer to the outline on the circuit board for proper orientation. The plastic base of the capacitor has slanted corners on one end, and these should match the pattern on the board. The black stripe on the top of the metal can of the capacitor should be closest to the outer edge of the circuit board.
3. Place the JST PH-series battery connector (Power PCB parts list POWER_PLUG) on its pattern. Follow the orientation of the footprint on the board. Current versions of the Power board also include a pair of through-hole terminals labeled +BAT and GND that could be used to attach wires from another battery connector type instead of the JST connector.
4. Place the voltage regulator (Power PCB parts list VREG) on its footprint.
5. Place 22 µF tantalum capacitor (Power PCB parts list C1) on footprint C1. The footprint for C1 has a white stripe at one end, and a + sign near that same end. The capacitor should have a white stripe on one end. Orient the stripe on the capacitor towards the + sign, so that it is closest to the microSD footprint.
6. Place the red LED (Power PCB parts list LED1) on footprint LED1. The footprint has a + sign marking the positive pad. When viewing the LED from above (normally requiring a magnifying lens), a green dot marks the ground pin of the LED. Place the LED so that its ground pin is on the pad opposite the pad marked with the + sign.
7. Place the green LED (Power PCB parts list LED2) on footprint LED2. As with the red LED, when viewing the LED from above, a green dot will mark the ground pin. The circuit board footprint has a + sign near the positive pin of the footprint. Place the LED so that the green dot is on the pad opposite the pad marked with a + sign.
8. Place 510 ohm resistors (Power PCB parts list R7, R8) on footprints R7, R8 (orientation does not matter).
9. Place the micro SD card holder (Power PCB parts list U$2) on its footprint.

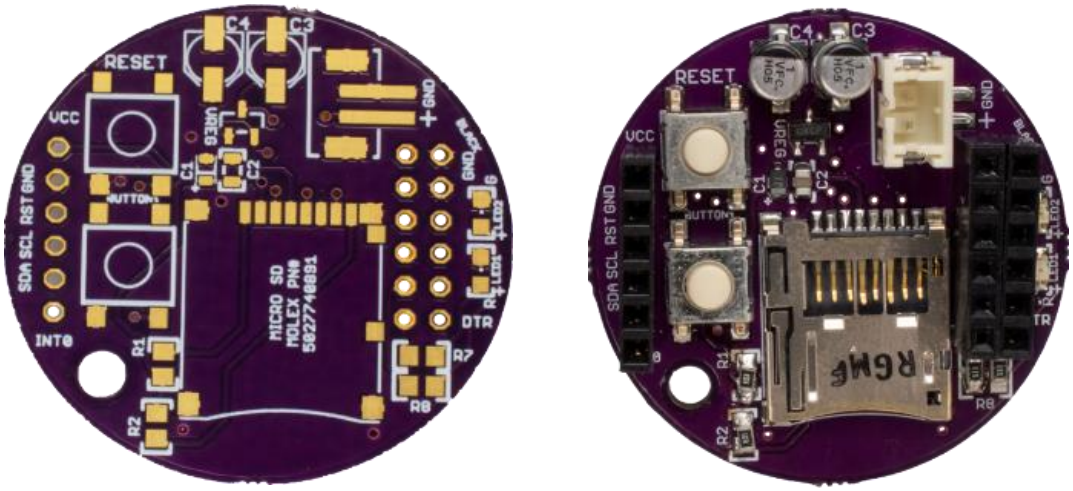


Figure 4. Left: The bottom surface of a bare power circuit board. Right: The bottom surface of a power circuit board with all components installed.

10. Place the pushbuttons (also known as tactile switches, Power PCB parts list RESET, BUTTON1). They will match up with the circuit board footprint in two orientations and will work in either orientation.
11. Place 100k ohm resistors (Power PCB parts list R1, R2) on footprints R1, R2.
12. Solder the surface mount components using a reflow solder oven or hot air reflow station.

Power circuit board assembly continued

13. Insert three 1x6 female wire-wrap headers (Power PCB parts list SPI_HDR, I2C_HDR, FTDI_HDR) from the bottom surface of the circuit board. Solder the pins on the top surface using a soldering iron. Leave the pins long so they mate with the female headers of the CPU circuit board.

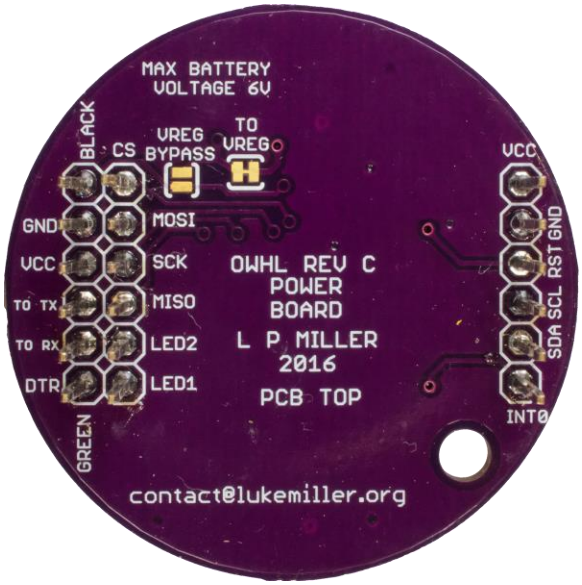


Figure 5. The top surface of a fully assembled Power circuit board. The pins of the 1x6 headers are left at full length after soldering.

Table 3. MS5803 circuit board parts list.

MS5803 PCB Schematic Label	Part Description	Manufacturer	Mfr. Part Number	Quantity	digikey.com PN	newark.com PN	mouser.com PN
SPI_HDR, I2C_HDR, FTDI_HDR	Female header (short tail for MS5803 disc)	Sullins Connector Solutions	PPTC061LFBN-RC	3	S7004-ND	BG095-06-A-N-D	
U1	MS5803-14BA pressure sensor	Measurement Specialities	MS5803-14BA	1	223-1626-5-ND	03AC1591	824-MS580314BA01-00
C5	Capacitor, surface mount 0805 0.1uF 10V ceramic	Taiyo Yuden	LMK212SD104KG-T	1	587-1134-1-ND	84R9733	963-LMK212SD104KG-T

MS5803 Sensor Board Build

1. The MS5803-14BA pressure sensor (MS5803 PCB parts list U1) comes pre-soldered, so no additional solder is needed to attach the sensor to the circuit board.
2. Apply liquid or paste flux to the 8 pads of the MS5803 footprint on the top surface of the circuit board, and place the MS5803-14 chip on the footprint.
    - Alignment: The MS5803-14 when viewed from above, has a blue dot in one corner of the chip (Figure 6 Center). This blue dot should be matched to the dot next to the MS5803 label on the circuit board. Additionally, the MS5803-14, when viewed from the bottom, has one longer solder pad. Match this longer solder pad to the one longer pad on the circuit board footprint (Figure 6 Left).
3. Using a reflow oven or hot air reflow station, solder the MS5803-14 to the circuit board.
4. After the MS5803-14 chip is attached, turn the circuit board over. Apply solder paste to the pads of position C5 labeled on the bottom surface of the board (Figure 6 Right). Place the 0.1 µF capacitor (MS5803 PCB parts list C5) in position C5.
5. Solder C5 using a hot air reflow tool or soldering iron.
6. Insert three 1x6 female headers (MS5803 PCB parts list SPI_HDR, I2C_HDR, FTDI_HDR) in the three 1x6 header positions on the circuit board. Insert the headers from the bottom side of the circuit board, so that their tails will be soldered on the top surface of the board. Solder using a soldering iron. Trim the tails on the top surface of the circuit board after soldering to avoid interference when the MS5803-14 is mounted to the bulkhead.



Figure 6. Left: The upper surface of a bare MS5803 sensor circuit board. Center: Top surface of the MS5803 board with MS5803-14BA sensor and 1x6 header pins installed. Right: Bottom surface of the MS5803 board with capacitor C5 and 1x6 headers installed.

Programming Newly Assembled OWHL Boards

Modifying boards.txt

We begin by creating an entry for the OWHL's hardware configuration. This is accomplished by modifying a file used by the Arduino software, `boards.txt`, which lists the configurations for the various Arduino boards.

Windows

- On a Windows installation, the file can be found in the Arduino installation folder (typically Programs (x86)/Arduino/).
- Navigate to hardware/arduino/avr/ to find `boards.txt`.

Mac OSX

- On a Mac OSX installation, open a Finder window and navigate to Applications > Arduino then long-click and choose "Show Package Contents".
- Then navigate to `Contents/Resources/Java/hardware/Arduino/avr` to find `boards.txt`.

The `boards.txt` file can be opened in a basic text editor (Notepad, TextEdit, etc.), and the following code (between the lines of ####) can be copied and pasted into `boards.txt`:

```
##############################################################
# For Arduino IDE v 1.6.3 or later
# Tested successfully using Arduino as ISP programmer to burn
# bootloader on OWHL.
# Cut and paste this entry into your existing boards.txt file
# found inside the /hardware/arduino/avr/ directory.
# Select the OWHL (8MHz internal clock) entry in the
# Arduino IDE menu under Tools>Board>

OWHLatmega328.name=OWHL (8 MHz internal clock)
OWHLatmega328.upload.tool=arduino:avrdude
OWHLatmega328.upload.protocol=arduino
OWHLatmega328.upload.maximum_size=30720
OWHLatmega328.upload.speed=57600

OWHLatmega328.bootloader.low_fuses=0xE2
OWHLatmega328.bootloader.high_fuses=0xDA
OWHLatmega328.bootloader.extended_fuses=0x05

OWHLatmega328.bootloader.file=atmega/ATmegaBOOT_168_atmega328_pro_8MHz.hex
OWHLatmega328.bootloader.unlock_bits=0x3F
OWHLatmega328.bootloader.lock_bits=0x0F
OWHLatmega328.bootloader.tool=arduino:avrdude
OWHLatmega328.build.mcu=atmega328p
OWHLatmega328.build.f_cpu=8000000L
OWHLatmega328.build.core=arduino
OWHLatmega328.build.variant=standard

##############################################################
```

After adding that code to the boards.txt file, and restarting the Arduino software, you should find an entry for “OWHL (8 MHz internal clock)” under the Tools>Board menu. Choose that entry when you burn the bootloader.

Burning the Bootloader

A brand new ATmega328p chip needs to have a small program (a bootloader) installed on the chip using an In-System Programmer (ISP) initially so that it can then be programmed and communicated with using a standard USB to Serial converter in later steps. There are several commercial ISPs available for Arduinos (**Figure 7**) but a standard Arduino Uno can also be used as an ISP. The instructions below outline the use of an Arduino Uno as an ISP.

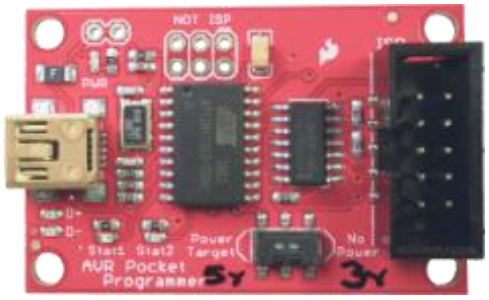


Figure 7. An example of an AVR Pocket Programmer that serves as an In-System Programmer, which is used to burn a bootloader program onto a newly assembled OWHL.

These steps are based on the instructions on this page: http://www.arduino.cc/en/Tutorial/ArduinoISP

- Open Arduino and set Tools>Board to Arduino Uno (this is the board you will use as the programmer).
- Upload the Example>ArduinoISP sketch to the Arduino Uno. The Uno is now programmed to behave as a programmer for new AVR chips (this is completely reversible).
- Place a 10µF capacitor between Reset and Ground on the Arduino Uno, with the negative leg in the Ground header.
- Change the Arduino Tools>Board to “OWHL (8 MHz internal clock)”
- Power down the Uno, and connect the ISP header on the OWHL to the appropriate pins on the Uno. The pin layout on the OWHL is shown in (**Figure 8**)

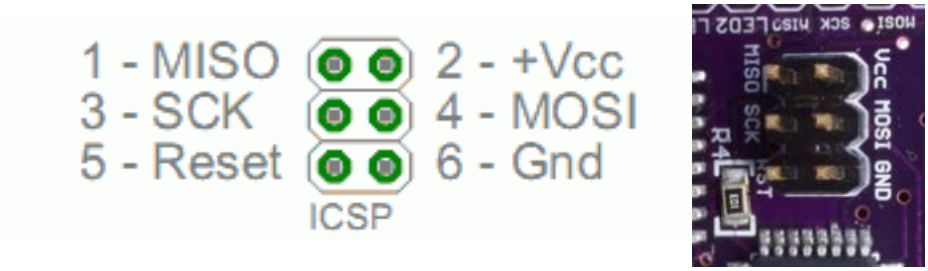


Figure 8. Left: connection diagram for the In-System Program header pins on the OWHL CPU board. Right: Close up photo of the ISP header on CPU board. Pin 1 (“MISO”) is located in the upper left of the header as pictured.

- Pin connections between the OWHL and Arduino Uno to allow the use of the Arduino Uno As ISP functionality:
    - OWHL Pin 1 MISO to Uno digital pin 12
    - OWHL Pin 2 +Vcc to Uno 3V3 pin
    - OWHL Pin 3 SCK to Uno digital pin 13
    - OWHL Pin 4 MOSI to Uno digital pin 11
    - OWHL Pin 5 Reset to Uno digital pin 10
    - OWHL Pin 6 GND to Uno GND pin
    - To be clear, the OWHL is the “slave” in this scenario, and the Arduino Uno is the “master” when burning the bootloader.

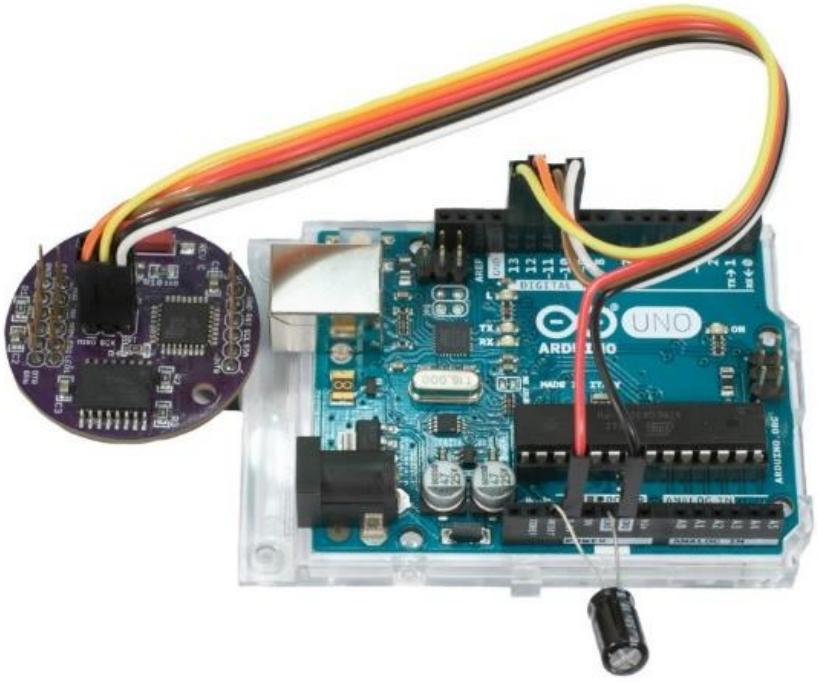


Figure 9. Using an Arduino Uno as ISP to burn the bootloader on a newly assembled OWHL CPU board. The OWHL board will be powered by the 3V3 and ground pins from the Arduino Uno in this setup.

- Plug the USB cable back into the Uno programmer board.
- Go to Tools>Programmer and choose “Arduino as ISP”. If you are using a commercial ISP board instead, choose the Programmer option appropriate for your programmer.
- Go to Tools>Burn Bootloader and watch the bootloader burn happen over a few minutes.

With a bootloader program installed on the OWHL, the ISP can be disconnected. The ISP should not be needed again for the OWHL.

Connecting the USB to Serial Adapter

After the bootloader has been burned onto a new OWHL, all further programming can use a USB to Serial adapter, also commonly called an FTDI adapter (FTDI is a manufacturer of electronic parts). These USB to Serial adapters can be sourced from several vendors, such as the Adafruit FTDI Friend or Sparkfun FTDI Basic (**Figure 10**).

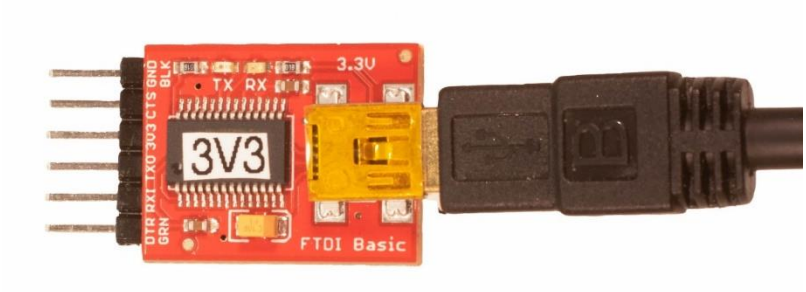


Figure 10. FTDI Basic adapter from Sparkfun.com, 3.3 volt model. A set of 6 pins (shown) is useful for attaching the FTDI Basic to the female FTDI header on the bottom of the CPU circuit board or Power circuit board.

The FTDI adapter can be connected to the 1x6 FTDI header, which is the outer header of the two rows of 1x6 headers found on the OWHL circuit boards. Each of the OWHL circuit boards has the FTDI header pins, so you can connect the FTDI adapter to either the CPU board directly, or the Power board when the CPU board is already attached to the Power board (this would be the normal arrangement when an OWHL is fully assembled).

The orientation of the FTDI adapter matters, and could be easily hooked up backward, so pay careful attention when connecting the FTDI adapter to the OWHL. One end of the 1x6 FTDI header on the OWHL boards is labeled “black” and “GND”, and the opposite end is labeled “green” and “DTR” (Figure 11). These labels correspond to the labels commonly used on various FTDI adapters and should be matched up when connecting the FTDI adapter to the OWHL header (Figure 12).

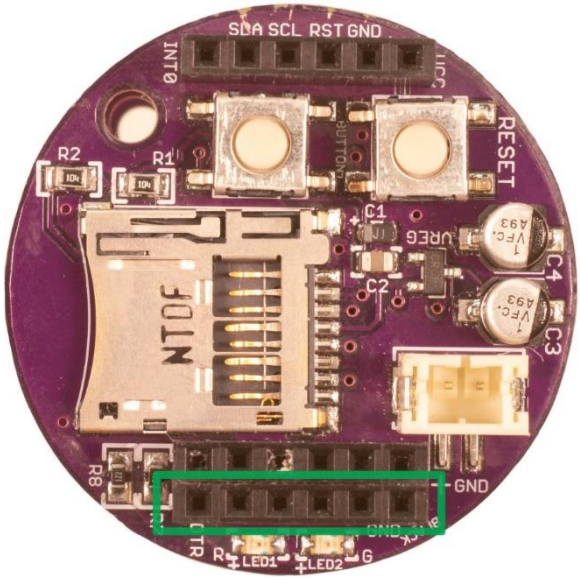


Figure 11. The green box highlights the FTDI header, viewed from the bottom surface of the Power circuit board. The left-most header pin is labeled ‘DTR’ and the right-most pin is labeled ‘GND’ and ‘black’.

When the FTDI adapter is connected via USB to your computer, and the Arduino program is open, go to Tools>Port and find the entry for your FTDI adapter.

- On a Windows computer, FTDI adapters will show up as a COM port (COM1, COM2, etc.).
- On Mac OSX, FTDI adapters will show with an entry similar to dev/cu.usbserial-XXXXXXX

Choose the entry for your FTDI adapter. If you're not sure which entry it is, try unplugging the USB cable from the FTDI adapter temporarily, look at Tools>Port and see which entry has disappeared and then reappears when you plug the FTDI adapter back into the USB port.



Figure 12. FTDI Basic adapter attached to the Power board of an assembled OWHL stack. The 6-pin terminal on the end of the FTDI Basic adapter has a pin labeled "DTR" on the left, and a pin labeled "GND" on the right, as pictured. These pins should be matched to the pins labeled "DTR" and "GND" respectively on the OWHL 1x6 female FTDI header on the bottom of the Power or CPU circuit boards.

Setting the Real Time Clock

With the USB to Serial adapter (FTDI adapter) connected and the FTDI port selected in the Arduino software, the OWHL will be powered up and ready to program using the Arduino software.

The first task will be to program the real-time clock (RTC), which should already have its CR1220 battery installed.

- In Arduino, go to File>Sketchbook>OWHL>settime_Serial
- The `settime_Serial.ino` program window will open in Arduino
- Click the button at the top of the Arduino window that looks like a right arrow "Upload", or go to Sketch>Upload
- The Arduino software will attempt to program the OWHL. You should see a notification in the lower section of the Arduino window telling you when the program is successfully uploaded.
- After a successful upload, open the Arduino Serial Monitor window, using the magnifying glass icon in the upper right of the window, or using the menu Tools>Serial Monitor
- In the Serial Monitor window at the bottom, you will find a menu item for the serial communications speed, which is typically set to "115200 baud" by default. Set this menu to "57600 baud" to communicate with the OWHL.
- The Serial Monitor should show a message prompting you to enter the date and time in the format YYYY MM DD HH MM SS

- The date and time can be typed in the text entry field at the top of the Serial Monitor window
- Follow the format exactly, including the spaces between values
- For example, if you want to set the clock to read 3:20:30 PM on January 6, 2019, you would type the values 2019 01 06 15 20 30
- Choose a time value in seconds that is slightly ahead of the current time as you type in the values
- When you have finished entering the date and time, wait until your clock time matches the value you entered, and hit the Return key to send the time value to the OWHL
- After a successful time upload, the Serial Monitor should start showing you the OWHL's current time.
- If you missed setting the correct value, you can retry as many times as you like by entering a new time in the Serial Monitor and hitting Return when that time ticks over.
- We **strongly recommend** always programming the OWHL clock in the UTC zone (also known as Greenwich Mean Time). This is a standard for most data collecting instruments and avoids any ambiguity about time zones or Daylight Savings Time changes. The OWHL is not capable of observing Daylight Savings Time.

To check that the Real Time Clock successfully keeps time on its own without external power, close the Serial Monitor and unplug the OWHL from the FTDI adapter. The OWHL will power down at this point, but the CR1220 backup battery on the Real Time Clock should keep the clock running. Plug the FTDI adapter back into the OWHL and open the Serial Monitor. The current date and time should be displayed on the serial monitor.

Programming a Serial Number Onto the OWHL

In cases where you will have multiple OWHL devices, it is desirable to make sure each OWHL imprints a unique serial number in the header of each output .csv file. To program a serial number, we use the `serial_number_generator.ino` program found in the Arduino File>Sketchbook>OWHL menu.

- Connect the FTDI adapter to the OWHL, and plug in the FTDI USB cable to the computer
- Bring the `serial_number_generator.ino` program window to the front
- Click on the Upload button (right arrow) found at the top of the Arduino window, or go to Sketch>Upload
- The upload status should be shown in the lower portion of the Arduino window
- The serial number generator program will write a serial number based on the upload date and time (for example SN20190612093030 if it was programmed on June 12, 2019 at 9:30:30AM) to a portion of the OWHL's memory called EEPROM memory. This memory is not normally accessed during OWHL operation and will retain the serial number even as you upload other programs to the OWHL
- If you open the Serial Monitor window, the programmed serial number should be displayed.

Programming the main OWHL data collection program

After the Real Time Clock has been set, and a serial number has been programmed, you can upload the main OWHL data collection program, which is the file `OWHL.ino`.

- Connect the FTDI adapter to the OWHL and plug the USB cable into the computer
- Open Arduino, and go to File>Sketchbook>OWHL>OWHL
- The `OWHL.ino` program should open
- Click the Upload icon (right arrow) at the upper left of the Arduino window, or go to Sketch>Upload
- The upload progress will be shown in the lower portion of the Arduino window
- After a successful upload, the OWHL should play a little chime to indicate that it is starting data collection. If there are errors during startup, including a bad date and time value or missing micro SD card, the red and green LEDs will flash and an annoying beep will sound.
- While the FTDI adapter is connected to the OWHL, the Serial Monitor window in Arduino can be opened to show the status of the OWHL.
- Now the OWHL will begin collecting data any time power is applied and will continue collecting data until power is disconnected.

Choosing a microSD Card

One of the main sources of power consumption in the OWHL is the microSD card, and identifying a power-efficient card can be important for getting the longest run time out of a set of batteries. As the drive to improve the battery life of consumer electronics has progressed, microSD cards have generally become more power efficient, and this is accomplished by having the card quickly go into a low power “sleep” state when it is not being actively accessed. Testing the power draw of an SD card, described below, is somewhat advanced or time consuming, so we will note that we have had good luck with name brand cards in the 4 to 32 GByte capacity range from major memory suppliers such as SanDisk, Kingston, and Samsung. The relatively simple electronics of the OWHL cannot use the extra features of extremely fast or high capacity microSD cards, so there is no advantage to buying more expensive cards advertised as being high performance. For those that want to estimate the efficiency of their cards, we detail two approaches we have used for monitoring power consumption.

Monitoring Battery Voltage

The simplest option for monitoring battery power usage, but also the most time-consuming, is to assemble a fully-programmed OWHL unit with a microSD card, attach a battery, and monitor the battery voltage over weeks or months. Using a handheld multimeter, measure the initial battery voltage at the start of the trial. A set of 3 new 1.5V alkaline cells should have a combined voltage of approximately 4.8 to 5 V, as brand new cells will often measure around 1.6 to 1.65 V. Make repeated recordings of the battery voltage approximately weekly to establish the trend in battery power consumption.

Figure 13 shows three example battery voltage curves for OWHL test units powered off of 3 “D” cell batteries. The test unit with an inefficient older unbranded microSD card depleted the batteries from an initial 4.8 V down to 3.81 V after 17 days and had reached the 3.1 V cutoff by 34 days. In the same 17 day time span, the two test units using modern SanDisk 8GB Class 4 microSD card had only dropped to 4.62 V from an initial voltage of 4.8 V, and by 34 days had dropped to 4.5 V, and both units ultimately ran for more than year before hitting the 3.1 V cutoff. The rate of discharge would be steeper for smaller battery cell capacities such as “AA” or “AAA” sizes, so **Figure 13** should not be used as a guide when using smaller battery sizes.

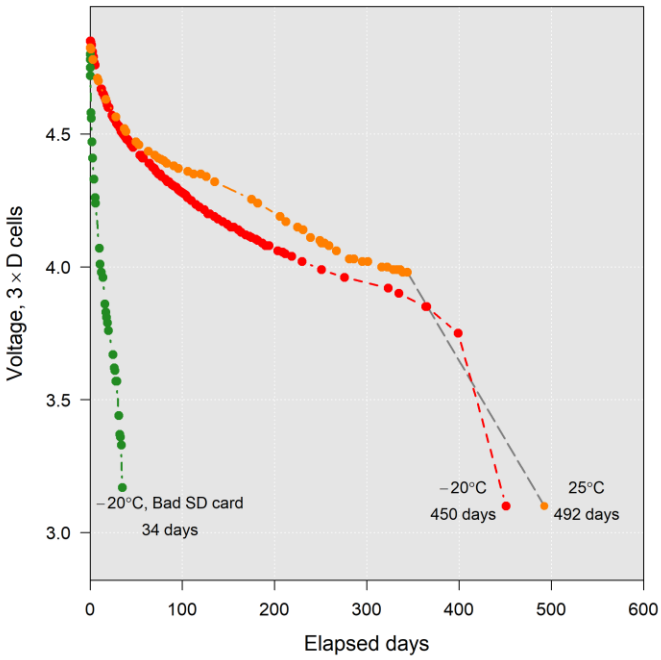


Figure 13. Example battery voltage depletion trends for three OWHL test units in the laboratory. All three devices were powered with 3 x “D” cell alkaline batteries, with an initial combined voltage of ~4.8V. The left most (green) curve illustrates the effect of using a microSD card that is not power efficient, while the two curves on the right used modern name brand SanDisk 16 GB cards purchased in 2014.

Monitoring Current Draw With an Oscilloscope

If you have access to an oscilloscope, it can be used to monitor the current draw of an assembled OWHL unit in real time. This method requires an oscilloscope, tools to cut, strip, and solder the wire used in the battery connector pigtail, and a 1 ohm axial resistor with a ¼ watt or higher power rating (**Figure 14A**). Axial resistors are the old-style resistors with two wire leads sticking out of a central ceramic cylinder, often with colored bands on the cylinder.

- Cut the negative (ground, typically black color) wire of the battery pigtail near its midpoint.
- Strip back about 5-10 mm of insulation from each half of the cut wire.
- Solder the 1 ohm resistor between the two halves of the cut and stripped wire (**Figure 14**B).
- Attach the oscilloscope probe lead to the end of the 1 ohm resistor closest to the OWHL.

- Attach the oscilloscope ground lead (often the short alligator clip hanging off of the oscilloscope probe) to the end of the 1 ohm resistor closest to the battery pack (**Figure 14C**).
- Plug the battery pack into the OWHL and let the OWHL start recording data.
- On the oscilloscope, set the Volts/Division to 10 mV, and set the time base (horizontal axis) to a long interval such as 250 ms or as high as 1 second. Set the oscilloscope to free run.
    - Alternatively, use the voltage trigger adjustment to set the trigger at a level about 15-20 mV above the baseline voltage reading seen on the oscilloscope screen while the system is running. This should catch and trigger on the high voltage peak that occurs at the start of a microSD card write event.
- While the oscilloscope is running, look for the occasional spike in the voltage trace, which indicates the increased current draw (converted to voltage by the oscilloscope and resistor) that occurs when the microSD card awakens and writes data to memory.

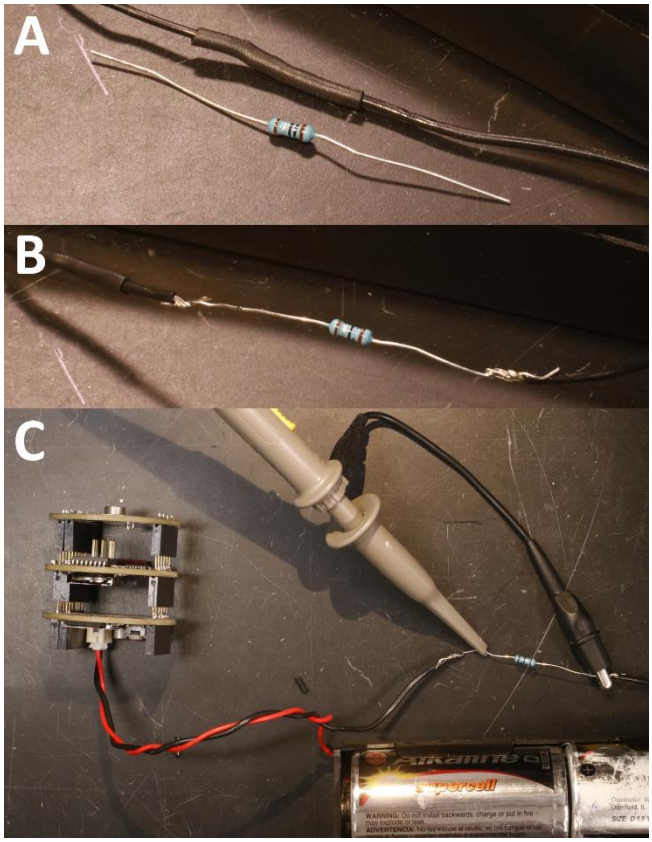


Figure 14. Inserting a 1 ohm resistor into the negative wire lead of the battery pack. A) Resistor and negative wire. B) The black wire is cut, stripped, and the 1 ohm resistor is soldered in place. C) A oscilloscope lead and ground clip are attached to the two legs of the 1 ohm resistor, and the battery pack is plugged into an OWHL programmed with the main data collection firmware.

The panels in **Figure 15** show a comparison of voltage traces from two different microSD cards mounted in an OWHL. The name brand (SanDisk, 16GB capacity) card shown in **Figure 15A** creates a voltage peak of approximately 60-75 ms duration during a write event, while the unbranded card used in **Figure 15B** creates a much longer duration voltage rise of around 300 ms during which the power consumption of the OWHL is substantially increased. The short duration rise of the SanDisk card will minimize the overall power consumption of the OWHL and give the longest run time.

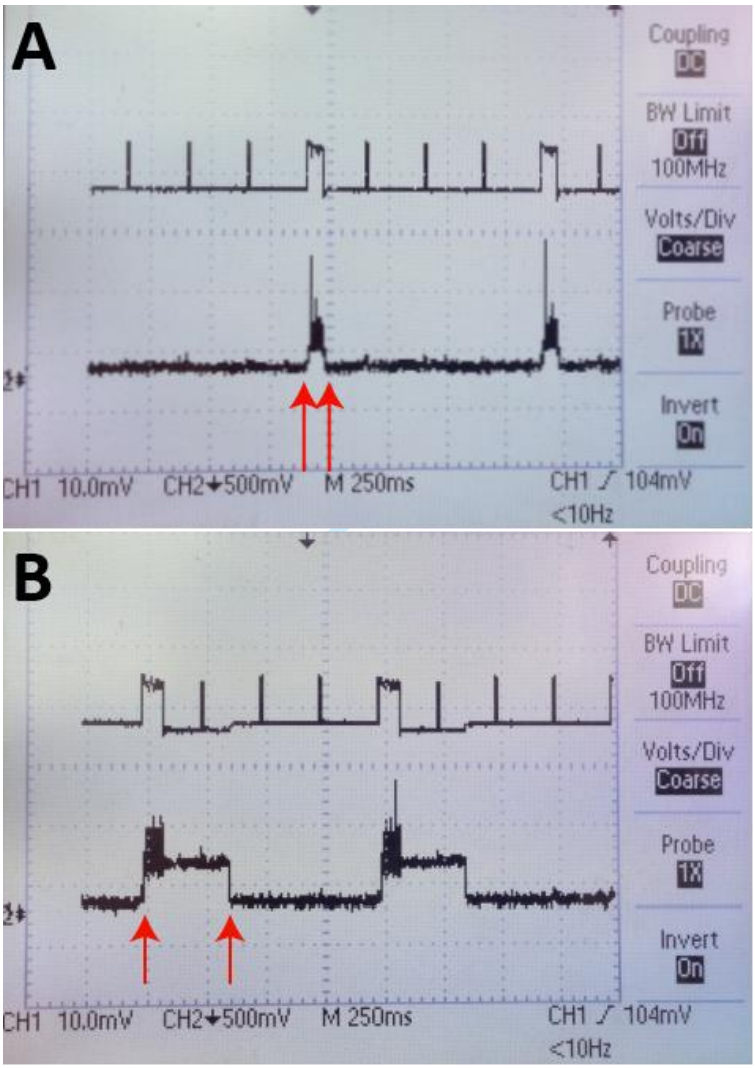


Figure 15. A) Voltage trace from an oscilloscope showing the rise in voltage measured across a 1 ohm resistor corresponding to the increased current draw during a microSD card write event. The red arrows mark the start and end of the write event, which lasts approximately 60-75 ms. The short duration of the peak for the microSD card write event illustrates that the card quickly returns to a low power sleep mode after the write event. B) Voltage trace from a low quality microSD card showing a longer period of high voltage (marked by the red arrows), corresponding to high current draw during and after a write event. Here the voltage trace stays high for approximately 300 ms before reentering a low power state.

Battery Holder Options

We provide two options for battery holders for the three "D-cell" batteries that run the OWHL. The first option, a nickel-plated steel holder (**Table 6** Item 23) is more expensive but does not require any modifications to fit inside the PVC pipe. The second option, a polypropylene plastic holder (**Table 6** Item 29), is too wide to fit in the PVC pipe with the batteries installed and needs to be modified by removing the corner edges of the holder as shown below. Both holders would typically need a JST connector pigtail soldered on to connect to the Power circuit board, although the current design of the Power circuit board now provides two through-hole terminals labeled +BAT and GND that could be used to solder wires for the user's own chosen connector style.

Attaching the JST Connector to the Metal Battery Holder

While the plastic holder comes with 26 ga wire leads, the metal battery holder does not. Short lengths of wire need to be soldered onto the ends of the metal battery holder as leads, which then are connected to the JST pigtail wire connector. The metal holder is approximately 18.5 cm (7.25 in.) in length. The additional wire with attached JST pigtail needs to be long enough to reach the JST connector on the Power circuit board. We found that an extra few centimeters of length helps to minimize pulling on the connector.

Tools Needed:

- Soldering iron & soldering materials
- Wire strippers
- Third-hand clips / Vise (or equivalent)



Figure 16. Detail of the terminal loop end of metal battery holder

Materials needed:

- Metal Battery Holder (**Table 6** Item 23)
- JST wire connector (**Table 6** Item 28)
- Additional wire (red, black) (**Table 6** Item 30)
- Heat-Shrink tubing (**Table 6** Item 31)

Attaching the Wire Leads

Each end of the metal battery holder has a small metal tabbed loop connector. A short section of stripped wire can be inserted into the hole and then twisted tightly to around the loop. This can then be soldered for strength.

- Cut a length of red and black wire
    - Additional length can be added to one wire lead to accommodate the length of the holder
- Strip approximately 1 cm (0.5 in) from both ends of each wire
- Insert the stripped end of one wire into the loop connector at one end of the holder
    - Bend the wire around the loop and then twist tightly to secure it
- Repeat with the other wire at the other end of the holder
- Flux and solder the twisted end of the wire to the loop connector

Attaching the JST Pigtail Connector

We will now solder on the JST pigtail (**Table 6** Item 28) to the ends of the wire from the battery holder. We recommend protecting the soldered connection with shrink wrap tubing. This should be installed before the connection is soldered

- Slide an appropriate sized shrink wrap tubing over the end of each wire lead coming from the battery holder.
    - Make sure that it is well away from the solder joint so as not to heat it accidentally
- Flux each end of lead wire to be soldered
- Flux each end of the JST pigtail connector wire
- Use a third-hand clamp, vise or equivalent to hold the wire connection to be soldered
- Solder the connection
- Slide shrink-wrap tubing back over the connection
    - Heat the shrink-wrap tubing to shrink it tightly and protect the solder connection

Modifying the Plastic Battery Holder to Fit Into the Housing

If cost is a concern, a polypropylene plastic holder (**Table 6** Item 29) can be purchased for nearly half the cost of the metal holder. However, it is too wide to fit in the PVC pipe with the batteries installed and needs to be modified by removing the corner edges of the holder as shown below.

Tools Needed:

- Utility Knife
- Belt or disc sander
- Soldering iron & soldering materials



Figure 17. Modified plastic three D-cell battery holder.

Materials needed:

- Battery Holder (**Table 6** Item 29)
- JST wire connector (**Table 6** Item 28)
- Additional wire (red, black)

A motorized belt or disk sander makes quick work of removing the edges from the battery holder.

- Remove the bottom edges of the battery holder (**Figure 17**)
    - Remove 1/8” to 1/4” of the corner at an angle of 45 degrees to the sides of the holder
    - Use a utility knife to clean up edges
- Using scissors or snips, remove the top 4 corners of the battery holder
- Install 3 D-cell batteries into the holder
    - Note that different brands of batteries have slightly different sizes
        - Duracell branded D-cells are slightly larger than EverReady branded cells
    - We secured our D-cell batteries in the plastic holder with electrical tape
        - This adds some slight thickness to the unit
- Test fit the holder with the batteries into a 1-1/2 in. PVC DWV Plain End Pipe section (**Table 4** Item 1)
    - Remove any additional edge via sanding, snips or utility knife as needed to fit
    - If the fit is too tight, it is very hard to remove the battery pack without damaging the attached wires



Figure 18. Modified 3 D-cell battery holder with JST connector pigtail soldered on.

See the previous section for attaching the JST pigtail connector to finish off the plastic battery holder assembly.

OWHL Housing Build

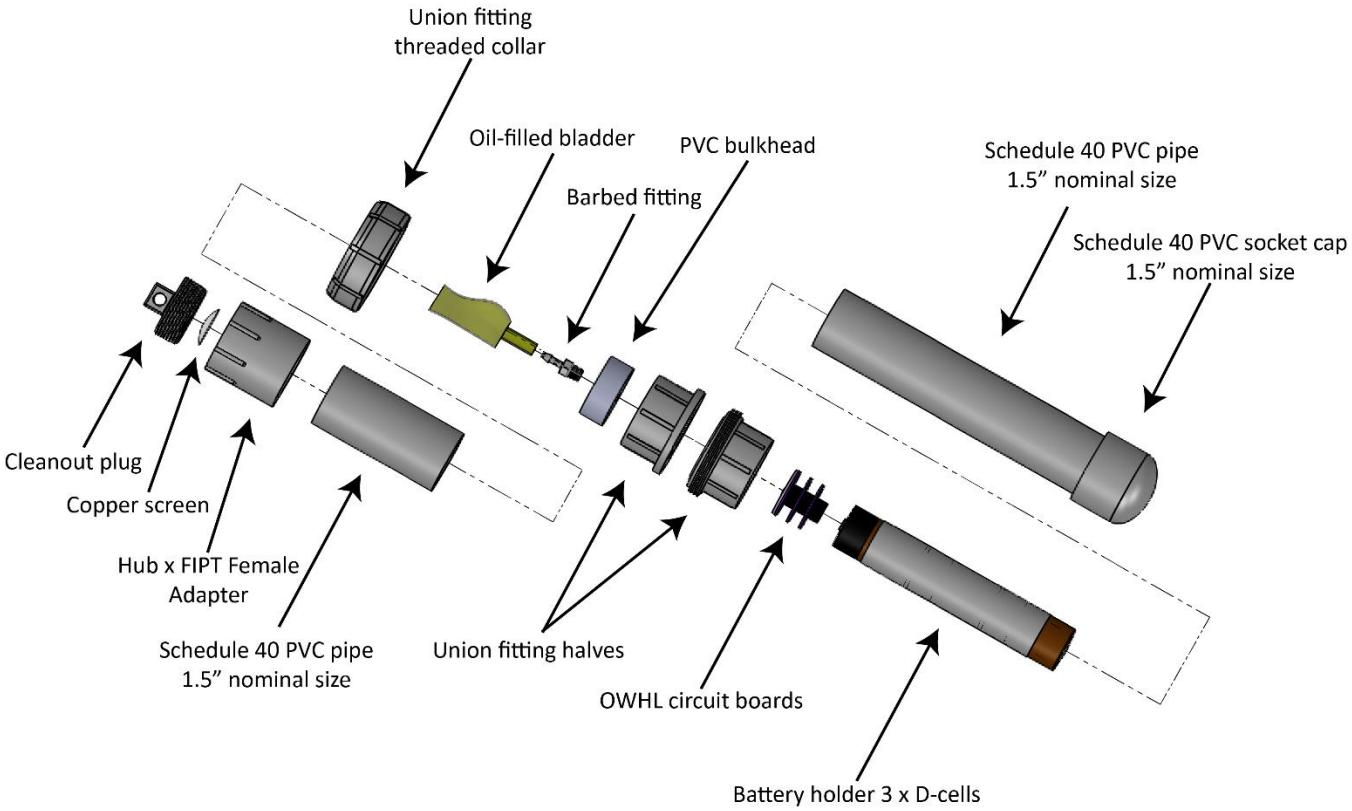


Figure 19. Exploded view of the OWHL housing components.

Making the Bulkhead

The bulkhead (**Figure 20**) is the interface between the Measurement Specialties MS5803-14BA pressure sensor (protected on the inside of housing) and an oil-filled reservoir (attached to the exposed outside of housing). It is most easily made by using a 2” (50 mm) hole saw on a PVC sheet to make a circular blank. The PVC blank is then drilled and threaded to make the bulkhead fitting for the logger.

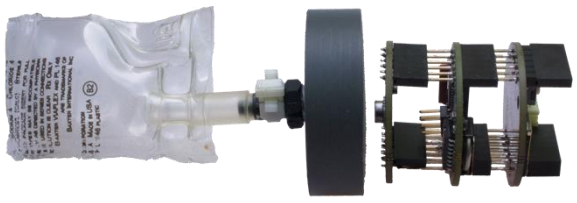


Figure 20. Bulkhead with oil reservoir and electronics.

Please note:

1. This part can also be produced using more highly specialized machine shop tools to produce a more uniform and cleaner part.
2. Our original design for the bulkhead included a small step in the piece that allows the electronics to sit flush with the union fitting. Some photos may show this detail however it is not required and frankly just makes more work.

Simple Build Instructions for Making the Bulkhead

This method to make the bulkhead fitting produces a less uniform and rougher part, but requires no highly specialized tools or training to produce. A 2" (5.08 cm) hole saw is used to plunge cut a flat 3/4" (1.9 cm) thickness PVC sheet stock. This method works best with a shop drill press

Tools Needed:

- Drill Press (preferred) or Powered Hand Drill
- Center Punch (optional)
- Hole Saw Arbor with 1/4" bit (6.35 mm)
- 2" Hole Saw (50mm)
- 'R' drill bit ('R' bit size = 0.3390" or 8.61 mm)
    - 11/32" is the closest fractional bit (0.3437" or 8.73 mm).
- Sandpaper (any rough grit of < 100 will work)

Materials needed:

- 3/4" thick PVC Sheet Material (e.g. McMaster-Carr #8747K117) (19.05 mm) (**Table 4** Item 8)

Cutting the Bulkhead Blank Using a Drill Press (or Powered Hand Drill)

- Measure out the placement for the bulkhead(s) on the PVC sheet.
    - While not required, using a center punch to mark the starting point for the drill bit keeps the bit from wandering and produces more uniform results
- Secure the sheet material with a clamp or vise to ensure repeatable placement of drill bit (**Figure 21A**). Do not attempt to hand-hold the sheet while drilling with a hole saw.
- Using a 2" (50.8 mm) hole saw on an arbor with a 0.25" (6.35 mm) bit:
    - Carefully cut a circular blank (the bulkhead) out of the PVC sheet.

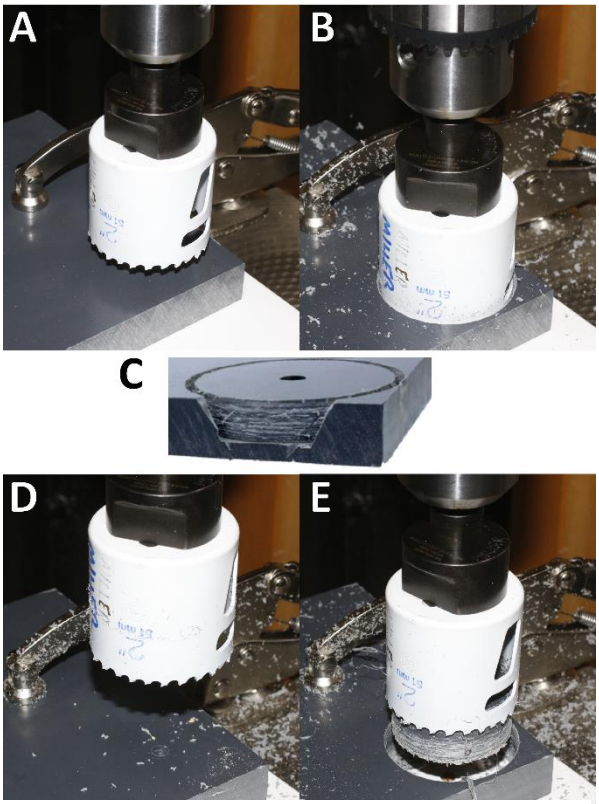


Figure 21. A) Clamp the PVC sheet securely in a drill press. B) Drill partway through the PVC sheet using a 2" (51mm) diameter hole saw with a ¼" center drill. C) A partially drilled PVC sheet. D) Turn the PVC sheet over, and center the hole saw using the center hole from the previous drilling step. E) Finish drilling through the PVC sheet, and remove the bulkhead, using a small flatblade screwdriver to pry it loose if it becomes stuck in the hole saw.

        - Not drilling completely through the sheet in one pass gave better results.
            - Drill to 90% of the thickness (**Figure 21B & C**)
            - Flip PVC sheet over
            - Use the exiting center hole to align the arbor bit (**Figure 21D**)
            - Cut remaining material (**Figure 21E**)
            - Remove PVC blank from the hole saw
    - Remove bulkhead from the hole saw
        - The newly cut blank needs to be pried out of the hole saw each time
            - A small flat-blade screwdriver works well

Drilling the Hole for the 1/8" NPT Fitting

Next step is to drill out a portion of the existing 1/4" (0.635 cm) hole in the blank for the 1/8" NPT thread barbed fitting on the "outside" of the bulkhead

Again, with a drill press or hand drill

- Use the 'R' drill bit (diameter 8.611 mm, 0.339") to make a pilot hole for the 1/8" NPT fitting
    - Drill ~ 1/2" to 5/8" (1.27 to 1.5875 cm) depth into one side of the bulkhead (**Figure 22**)

Continue to "Tapping the hole for the 1/8" NPT fitting"



Figure 22. Left: Use a size R drill bit to partially drill the existing center hole in the bulkhead. Right: The drilled hole in the bulkhead for the barbed fitting. This hole will have threads cut into it using a tap.

Tapping the Hole for the 1/8” NPT Fitting

The larger hole needs to have threads cut into it (tapped) to accept the fitting that will attach the oil reservoir (**Figure 22**). Tapping plastic can be tricky as it is easy to either misalign the tap and have it enter at an angle, or the new threads can break if too much force is exerted. Make sure that the tap is started in the hole as close to vertical as possible. Hand starting the tap is acceptable if done cautiously, however, starting the tap with the aid of a jig, a drill press or with a specialized tapping tool found in many machine shops is preferable. Only cut a few threads at a time, backing out to clear the cut material and then repeating. Never force the tap, especially as you reach the bottom of the hole, as it can easily break the newly cut threads. Using a plastics safe cutting lubricant can also help.

Tools Needed:

- Hand tap wrench
- 1/8” NPT tap (tapered pipe thread)
- Vise (or equivalent)

Materials needed:

- PVC Bulkhead
- Plastics safe cutting lubricant (optional)

Steps

- Gently secure the bulkhead in a vise or equivalent, as cutting threads is hard to do by hand-holding the material (**Figure 23**).
- Carefully begin by aligning the tap with the axis of the hole.
    - If using a cutting lubricant, apply a few drops to the tap first
- Using gentle pressure, begin by turning the tap in a clockwise direction.
    - Make slow clockwise turns until you feel the tap beginning to engage the PVC and cut new threads.
    - After cutting the first few threads, gently reverse direction and clear out the cut material.
- Carefully realign the tap and gently turn clockwise again to re-engage threads at the previous endpoint.
    - Continue cutting new threads a few at a time and then backing out to clear cut material until you reach the desired depth of ~ 3/8” (0.9525 cm) of the threaded hole.
    - Use cutting lubricant as needed



Figure 23. Left: Align the tap in the hole, ensuring that the tap stands perpendicular to the bulkhead. Slowly turn the tap in clockwise a few turns at a time by hand, and back the tap out counterclockwise every few turns to clear out the cut material. Right: The tapped threads are visible center hole, next to the barbed fitting.

Sanding the bulkhead surface for gluing the sensor board to it

The smooth PVC on the sensor side of the bulkhead needs to be roughened up to give better adherence for the glue (**Table 7** Item 31).

- Use some rough grit sandpaper to roughen the sensor-side surface of the bulkhead (**Figure 24**).

Gluing the MS5803 Circuit Board to the Bulkhead

Although the MS5803 circuit board has a series of holes that can be used for mounting the board against the bulkhead, we have found that adhesive marine sealant is sufficient for adhering and waterproofing the interface between the circuit board and the PVC bulkhead.



Figure 24. Roughen the interior surface of the bulkhead where the pressure sensor circuit board will be glued.

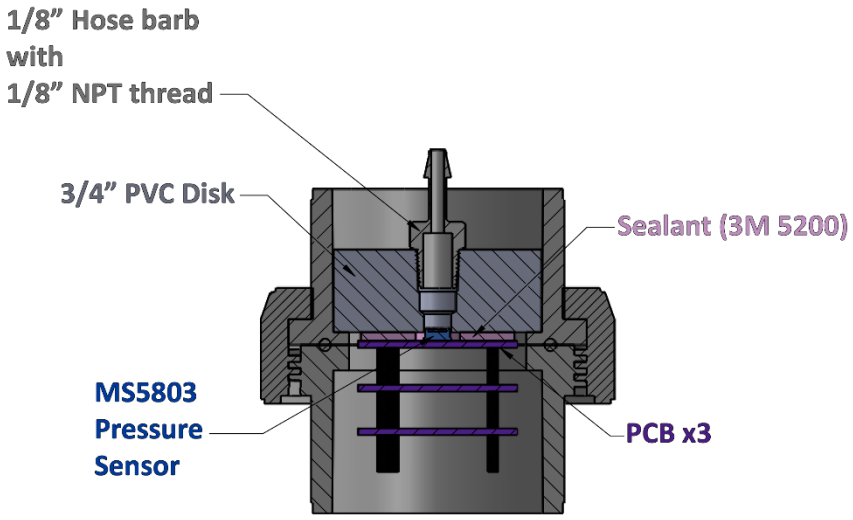


Figure 25. Cross-section view of the union fitting. The OWHL pressure sensor board is glued to the bottom of the ¾” PVC bulkhead fitting with 3M 5200 adhesive, and the bulkhead fitting is glued into the upper half of the union fitting.

Tools needed:

- Spring or screw-clamps

Materials needed:

- Assembled MS5803 circuit board
- PVC bulkhead (from the previous section)
- 3M Marine Adhesive Sealant Fast Cure 5200

Steps

- Apply a thin layer of 3M 5200 sealant to the PVC bulkhead on the side with the smaller hole
- Apply a thin layer of 3M 5200 sealant to the top surface of the MS5803 circuit board. (**Figure 26**)

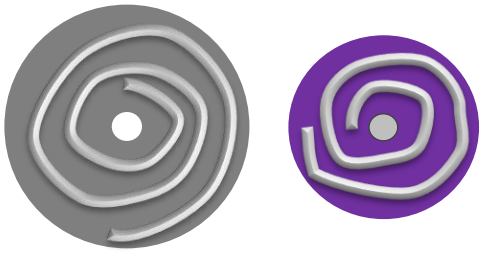


Figure 26. Left: apply adhesive to the surface of the bulkhead fitting on the side with the smaller hole. Right: apply adhesive to the top surface of the MS5803 circuit board, taking care to avoid getting adhesive on the white membrane of the sensor itself.

- Press the adhesive-covered surfaces of the bulkhead and circuit board together. Use a gentle twisting motion to try to eliminate air pockets and spread the adhesive evenly. Ensure that the adhesive does not get on to the white membrane of the MS5803 pressure sensor as it is inserted into the hole of the bulkhead, but do make sure to get an even layer of adhesive between the inner surface of the hole in the PVC bulkhead and the stainless steel ring of the MS5803-14 sensor (**Figure 27**)

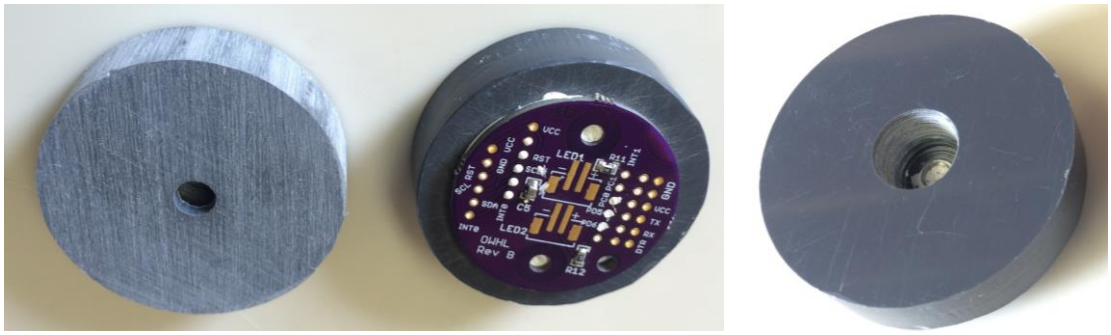


Figure 27. Left: roughened PVC bulkhead ready to receive adhesive. Center: MS5803 circuit board adhered to the bulkhead. Right: A view into the larger hole of the PVC bulkhead after mounting the MS5803 circuit board. The white membrane of the MS5803-14BA pressure sensor is visible in the center, and a layer of white adhesive has filled the space between the stainless steel ring of the pressure sensor and the PVC bulkhead hole.

- Use a pair of spring clamps or c-clamps to apply light pressure to the two pieces and hold them together while the adhesive cures for 24 h (**Figure 28**).



Figure 28. A pair of c-clamps maintain light pressure on the MS5803 circuit board and the PVC bulkhead while the adhesive cures.

Modifying the Cleanout Plug

We need to drill out the end of the cleanout plug to allow for an unimpeded flow of water around the oil reservoir. These holes also provide an anchor point for standard wire-ties. A copper mesh screen is glued in place to reduce fouling and prevent small fish from making it a home (**Figure 29**).

Tools Needed:

- Snips or scissors
- 1/4” to 1/2” Step bit (or equivalent) to drill plastic plug
- Vise (or equivalent)

Materials needed:

- Copper Mesh
- 1-1/2 in. PVC DWV MIPT Cleanout Plug (**Table 4** Item 5)
- 5-minute epoxy or epoxy putty

- Cut a 2.5 cm x 2.5 cm (1” x 1”) square of copper mesh (**Table 4** Item 9)
- Drill a 3/8” hole on each side of the square end
- On the inside of cleanout plug
    - Epoxy in a 1” x 1” piece of copper wire mesh to prevent fouling and fish entry



Figure 29. Views of the modified cleanout plug. The image on the right shows the copper mesh screen epoxied into place inside the plug.

Making the Oil Reservoir

The Measurement Specialties MS5803-14BA pressure sensor is not rated for continuous saltwater use and given its small size, can be easily fouled. To mitigate this, our design isolates the pressure sensor port from direct seawater contact, using an oil-filled bladder as an intermediate connection between the outside seawater and the surface of the pressure sensor. The bladder is made by modifying a small 25mL intravenous fluids (IV) bag once used for saline (**Figure 30**).
* Note: any IV bag size can be used

A commercial "Impulse Heat Sealer" like those used to package small retail products was used to carefully fuse the IV bag in several key places (**Figure 31**). Since the IV bag has two ports, we can get 2 oil reservoirs from each IV bag.

Tools Needed:

- Impulse Heat Sealer
- Utility Knife or Scissors
- 3/16" Drill Bit or equivalent (4.5 mm, #13 or 0.1875")

Materials needed:

- 25 mL Saline IV bags

Items needed to test bag:

- Aquarium airline tubing (needs to fit directly onto IV bag nipple or use with an adapter)
- Cup (filled with water)



Figure 30. Saline IV bag used to produce the oil reservoir.

Reduce the size of the IV bag and drain out the saline solution

- With scissors, cut the top off the IV bag first to drain it
    - Our finished oil bags have an approximate length of 6 cm as measured from the bottom of the bag (tube end)
    - Drain out the Saline
- Remove the protective caps from both tubes
- Using a 3/16" (0.476 cm) drill bit:
    - Cut out the internal seals in both the tubes
- With scissors, cut the IV bag in half, lengthwise
- Let the bag dry before the next step
    - The bags need to be reasonably dry inside as any saline left in the bag may prevent the heat sealer from working well

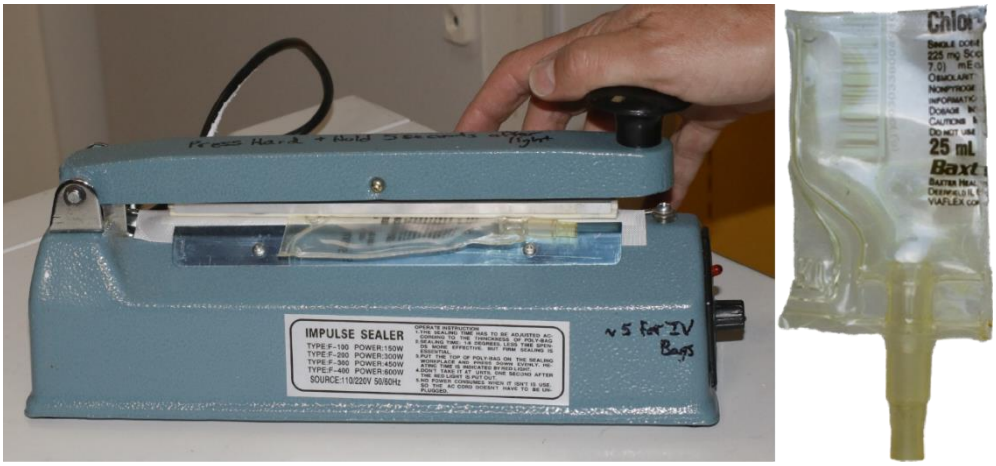


Figure 31. Left: Using a heat sealer to seal the cut edge of an IV bag. Right: A modified IV bag, after re-sealing the trimmed edges of the cut-down bag.

Using the Impulse sealer

- To seal the bag edges
    - Set the dial on the impulse sealer for the max of 8 seconds.
    - Use the impulse sealer to heat each side of the bag and fuse the melting plastic
        - Heat the bag approximately 2-3 times on each side of the bag.
            - Push and hold the sealer for the 8 seconds, three times in quick succession
            - Quickly flip the bag over and use the sealer an additional 2 to 4 more times (depending on how hot your sealer is)
                - For the last 2-4 presses, just ever so slightly move the bag from side to side to broaden the contact patch.
            - At the end of the last 8 seconds press, push down hard on the impulse sealer bar, holding this for an additional 10-15 seconds with the bar closed
                - Let the newly melted seam cool slightly with the bar pressure on it before removing it from the sealer.

Test the oil reservoir for any leaking

- To test the new edge seals of the oil reservoir bags:
    - Fit an aquarium airline tube to the bag nipple
    - Using a container of water, gently blow on the airline tube and look for any air leaking from the reservoir bag.

For any leaks found:

- Repeat the above steps as necessary directly on the leak point
    - Sometimes the leak is right at the corner of the two intersecting heat points
        - In this case, reseal at 45 degrees to the corner, making a small sealed triangle point over the leak

Housing Assembly

Note: Our design includes a small section (~ 4" length) of clear PVC (**Table 4** Item 7) in the housing to be able to see any leaking during deployment and to provide piece-of-mind in the setup, operation, and recovery of the OWHL (**Figure 32**). This is not critical to the design and does add a few dollars in cost to each housing.

Tools Needed:

- PVC Saw (or equivalent to cut the PVC pipe to length)
- Crescent Wrench (or equivalent for the 1/8" NPT fitting)
- Pliers (Needle Nose Pliers recommended)
- A syringe with large bore tip
- A syringe with fine bore tip

Materials needed (pipe sizes are given as nominal diameters):

- 1-1/2 in. PVC Socket Cap (**Table 4** Item 2)
- 1-1/2 in. PVC DWV Plain End Pipe (**Table 4** Item 1)
- 1-1/2 in. PVC Slip x Slip Union (**Table 4** Item 3)
- 1-1/2 in. PVC DWV Hub x FIPT Female Adapter (**Table 4** Item 4)
- 1-1/2 in. PVC DWV MIPT Cleanout Plug (**Table 4** Item 5)
- 1-1/2 in. PVC S x S Coupling *(Optional)* (**Table 4**, Item 6)
- 1-1/2 in. Clear PVC Pipe *(Optional)* (**Table 4**, Item 7)
- PVC Primer (**Table 4** Item 10)
- PVC Solvent Cement (**Table 4** Item 10)
- Mineral Oil

*Cut List to Make *One* OWHL Housing*

1. 1 x 4" piece of 1-1/2" DWV plain end pipe
2. 1 x 10" piece of 1-1/2" DWV plain end pipe

   (** if using optional clear section add the following:)
   - 1 x 4" piece of 1-1/2" Clear PVC tube
   - 1 x 6" piece of 1-1/2" DWV plain end pipe

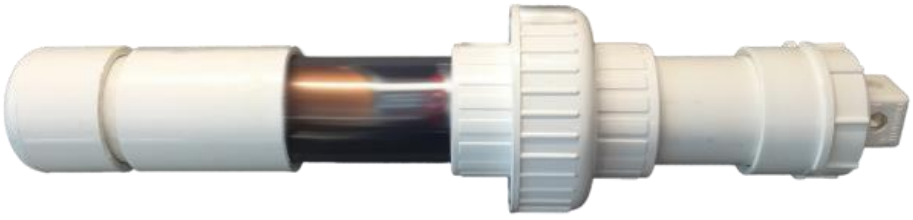


Figure 32. Assembled OWHL housing.

Assembly of the Housing Parts in Order

1. End cap
2. 10” length of 1-1/2” PVC Pipe (6” length if using optional clear section)
3. S x S Coupling *(Optional)*
4. 4” piece of 1-1/2” Clear PVC tube *(Optional)*
5. Bulkhead fitting
6. Union – Threaded half
7. Union O-ring (carefully set aside the union fitting O-ring for final assembly)
8. Union – Flange half
9. Union – Locking nut
10. 4” length of 1-1/2” PVC pipe
11. Hub x FIPT Female Adapter
12. MIPT Cleanout Plug

For all PVC solvent cement welds (gluing parts together)

- All pieces to be glued should be reasonably clean and debris free
- All glue joints should be primed with Purple PVC Primer or equivalent
- Both pieces to be glued should get a light coating of PVC solvent cement at the mating points using the dauber brush provided in the cap of the PVC cement
- Pieces should then be pushed together tightly and a slight twist given to spread the glue
- Hold the pieces together for at least 30 to 60 seconds after gluing
    - The heat from the solvent cement weld reaction can cause pieces to back out from expansion.
- Begin by gluing the bulkhead fitting to the threaded half of the union fitting
    - Make sure that the 1/8” NPT threaded fitting side is facing out
    - Coat the area that the bulkhead will sit in the union fitting with PVC solvent cement
    - Insert the bulkhead and twist slightly to spread the solvent cement
    - Use a paper towel to clean up any excess cement in the union fitting
        - The 4” long piece of pipe will need to be glued in above this point
        - Any dried PVC cement may prevent it from seating properly
- Glue the End cap on to the 10” length of pipe
    - (6” length if using the clear section)
- Glue the 10” length of pipe into the threaded end of the Union fitting above the bulkhead
    - (If using the clear pipe section, glue on the coupling and then clear piece of PVC)

Oil Reservoir Installation

Install the oil reservoir components from the previous section.

- Install the 1/8” NPT fitting using 3 wraps of Teflon thread tape or equivalent
    - Use a crescent wrench or equivalent to tighten the fitting
        - Do not over tighten
- Install the 4” length of pipe to protect the oil reservoir
- Install the FIPT Female Adapter

Filling the Oil Reservoir

It is now easiest to fill the oil reservoir with mineral oil before screwing on the cleanout plug. The end goal is a partially oil-filled bag attached to the barbed nipple fitting with no air bubbles in it (or only a few tiny bubbles less than 1 mm in diameter). The bag should only be half-filled with mineral oil to avoid introducing extra pressure on the sensor due to the elasticity of a fully-filled IV bag.

- Use a syringe with a large-bore end to fill the oil reservoir with mineral oil
    - Add mineral oil to the IV bag
        - Start by pressing the bag flat to push out most of the air
        - Hold the bag upright with the tube fitting up
        - Fill the bag about half full of oil
            - There may be some air bubbles remaining in the bag
                - Gently tapping the bag may help get the larger bubbles out
                - Squeezing carefully from the bottom of the bag may help force air bubble out the top of the tube fitting
            - Use a fine-gauge syringe to suck out any remaining smaller air bubbles
    - Sit the bulkhead with the barbed fitting upright. Add mineral oil to the barbed pipe fitting until a small meniscus of oil forms at the top of the fitting
        - Tap the fitting and bulkhead repeatedly to make sure any bubbles trapped in the bottom of the barbed fitting near the sensor port rise to the top and escape
        - A long syringe needle can be used to inject oil deep in the barbed fitting and agitate the fluid to force air bubbles up out of the barbed fitting
    - Install the oil reservoir bag on to the 1/8” NPT fitting barbed nipple. Both parts should be brimming with oil, and a bit of oil will escape as you press them together, hopefully without introducing any large air bubbles.
    - Install the clamp on to the oil reservoir tube over the barbed fitting
        - About halfway down the tube and between the barbs gives the tightest seal
    - Squeeze clamp with pliers carefully to seal
        - Over tightening the clamp can lead to it breaking

Once the bag is mounted on the barbed fitting, test the bag integrity by pressing down with the heel of your hand on the assembled oil-filled bag against a hard surface like a tabletop or lab workbench. The oil reservoir should not pop or leak.



Figure 33. An oil reservoir bag attached to the barbed fitting that mounts in the bulkhead.

Calibration

After the OWHL electronics and housing have been assembled, the device can be statically calibrated if desired. Calibration could be accomplished by lowering the OWHL into a swimming pool or other still body of water (with minimal surface wave action) to a set of measured depths, and tracking the time each depth was reached. Alternatively, we have mounted an OWHL in a port at the base of a vertical plastic pipe which can be filled with water to known depths, again recording the time that each known depth of water above the sensor was achieved. The timestamps in the CSV file produced by the OWHL can then be used to extract pressure readings at times that corresponded to the OWHL being held with a static head of water above it. All readings will include the pressure due to local air pressure, so we subtract off the initial pressure reading when the OWHL is not submerged from all of the test pressure readings. When using freshwater for calibration, we converted the reported pressure in mbar to meters of freshwater by dividing by 98.1. Using our standpipe calibration apparatus, we obtained excellent fit between the known depth and observed depth for an OWHL unit shown in **Figure 34**.

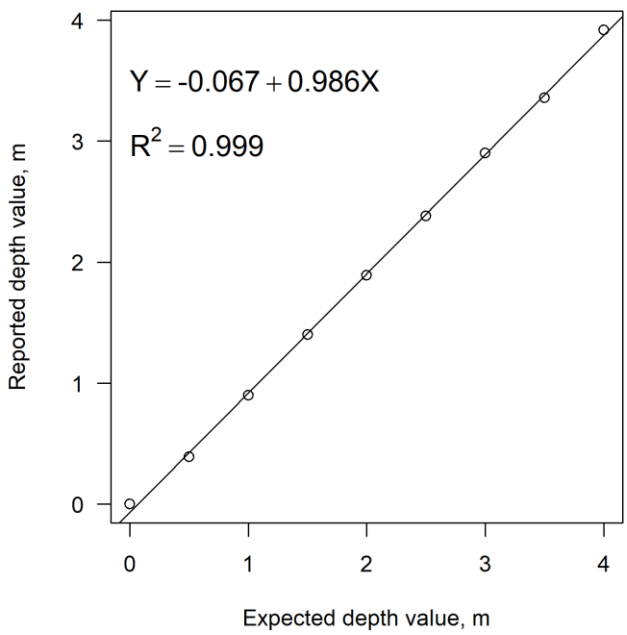


Figure 34. Calculated water depth values reported by an OWHL versus known water depths in a calibration apparatus. The regression equation and R^2 for the best fit line between the expected and reported depth values is shown.

Anchoring Methods

The primary concern for anchoring an OWHL is that the device be held in a fixed position relative to the sea floor. It should not be attached to a subsurface float or other structure that would let the OWHL change position in the water column during the deployment. In relatively calm conditions, attaching the OWHL to a lead brick resting on the bottom could be sufficient (**Figure 35**).



Figure 35. An OWHL attached to an 11 kg (25 lb) lead brick with band clamps. (Photo credit: The Bay Foundation)

Anchoring In Soft Sediments

In mud or sand-bottom areas, we have successfully used sand anchors as a mounting structure for OWHLs. Two or more sand anchors and stainless steel worm-drive band clamps can provide a rigid attachment (**Figure 36**). Sand anchors, also known as ground anchors or earth augers, are available from various online retailers. Our anchors are constructed from 1/2" steel, with a 3-inch wide helix forming the auger that screws into the sediment. Stainless steel band clamps can be purchased in a wide variety of lengths to accommodate housings or protective pipes of different diameters. Be sure to source worm drive band clamps that use a stainless steel screw, since many band clamps advertised as stainless steel often contain a non-stainless steel worm drive screw.

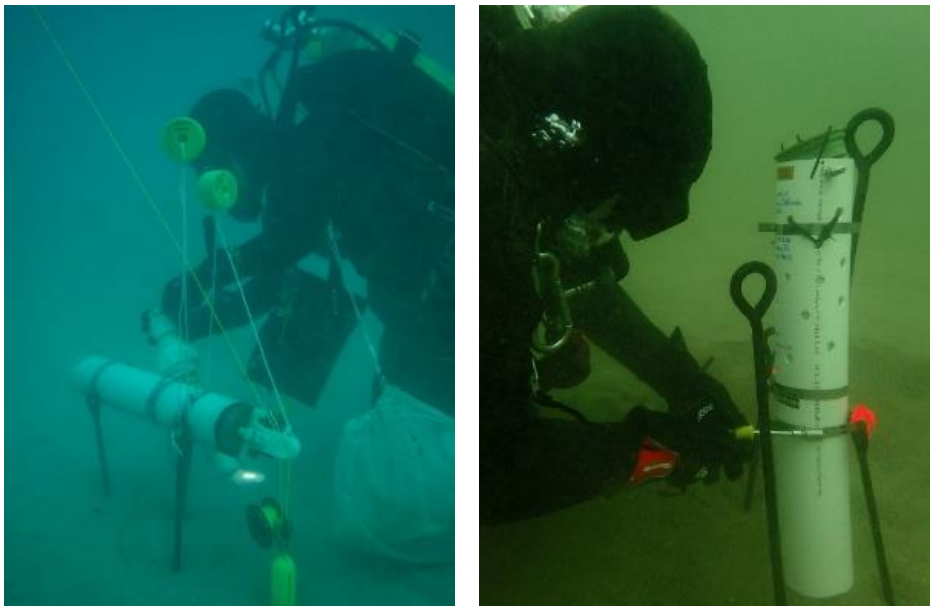


Figure 36. Using sand anchors to secure sensors in soft sediments. (Photo credit, left - Aaron Ninokawa, right – The Bay Foundation)

Anchoring in Hard Reef

In hard bottom situations, it may be desirable to place permanent mounting points in the reef itself. We have used a pneumatic drill (1/2" Reversible Air Drill, SKU 98896, Central Pneumatic, HarborFreight.com) and 0.9525 cm (3/8") diameter masonry drill bit to drill holes in rock reef. The air drill uses a 1/4"-18NPT threaded inlet which is standard for many air tools, but this fitting requires an adapter to attach to common SCUBA 1st stage regulator low-pressure ports, which use a 3/8"-24 UNF thread. These adapters are commonly sold as "NPT to SCUBA adapters". After drilling the holes, we anchor stainless steel eye bolts in the hole using two-part marine epoxy (A788 Splash Zone Epoxy, Pettit, Rockaway, New Jersey, USA). Stainless steel worm-drive band clamps can then be used to anchor the OWHL housing to the eye bolts while being sure to confirm that the OWHL will not rock or shift once fastened down (**Figure 37**).



Figure 37. Left: Stainless steel eye bolts epoxied into holes drilled in a rocky reef. Right: OWHL secured to an eye bolt using a stainless steel band clamp.
(Photo credit: The Bay Foundation)

Anchor Block Build

In rocky, mixed substrate deployments in the Gulf of Maine, we have successfully used a heavy concrete "deck block" that has been modified to fit a 4" diameter piece of PVC pipe. This pipe is then used as a protective mounting structure for the OWHL, allowing us to quickly remove and replace the OWHL for each deployment. The deck block is trapezoidal and weighs approximately 19 kg dry. The 4" PVC pipe is anchored to the block via concrete screws and a stainless steel washer. Our later design included an end cap to facilitate faster deployment of the OWHL in the field. The attached eye bolts provide a robust anchor point for a marker float and for using a lift bag (**Figure 38**).



Figure 38. Left: Anchor block showing PVC pipe and SS lifting eye-bolt. Right: Anchor block with lifting line and subsurface buoy line attached.

We have had good luck placing these blocks by lowering them from a boat with a line and then using a lift bag to move it to the final desired position. By placing the block in a slightly protected location such as under an overhang of a larger boulder or wedged into an outcropping crack, they have lasted for many field seasons.

Materials needed:

- Deck Block
- 4” PVC Plain End Pipe (nominal diameter) (**Table 8** Item 35)
- 4” PVC End Cap (nominal diameter)
- 5/16” thread x 4” length Stainless Steel Eye Bolt *(or equivalent)* (x2)
- 5/16” thread Concrete Anchor/Lag Shield (x2)
- 1/4” thread x 2.5” length Concrete Anchor Screws (x2)
- 1/4" Stainless Steel Washer (x2)

Tools Needed:

- PVC Saw (or equivalent)
- 3/4” (1.9 cm) Concrete Drill Bit
- 3/16” (0.47625 cm) Concrete Drill Bit
- Hole Saw Arbor with 1/4” (0.635 cm) bit
- 2” (5.08 cm) Hole Saw
- 5/16” nut driver drill bit
- 1/2” (1.27 cm) general-purpose twist drill bit



Figure 39. An OWHL mounted inside 4 inch diameter pipe on a concrete anchor block.

- Drill (Hammer drill preferred)
- Angle Grinder *(or equivalent)*
- Masonry blade
- Hammer or Mallet
- Concrete cold chisel

Deck Block Modification

The top of the block must be cut to accept the 4” diameter pipe (**Figure 40**).

- Using the masonry blade on an angle grinder
    - Cut off approximately 1” (2.54 cm) from each side of one of the top channels *(the overlap from the edge)*
        - The masonry blade we used was for a 4.5” grinder and makes a 2” deep cut)
    - Use a cold chisel and hammer to break out the remaining piece of concrete

Now we will make the holes for the lag shields/concrete anchors

- Using the 3/4” (1.9 cm) masonry drill bit
    - Drill two 3/4” (1.9 cm) diameter x 2” (5.08 cm) deep holes approximately 2” (5.08 cm) from the edge on each side of the other channel on the top of the block



Figure 40. The modified upper surface of the anchor block.

PVC pipe modification

We cut several holes into the 4” diameter x 16” long PVC pipe to facilitate driving the concrete anchor screws in and to allow a line to pass through the pipe if needed.

- Cut a length of 4” PVC DWV pipe to 16” (40.6 cm) length
- Find the center of the pipe
    - Drill two 2” (5.08 cm) diameter holes at the center point opposing each other 180 deg.
    - Rotate pipe 90 degrees
    - Drill two 2” (5.08 cm) holes approximately 2.5” to 3” (6.35 to 7.62 cm) on either side of the center point of the pipe
- Glue on the PVC end cap onto one end of the pipe
- On the opposite end of the pipe:
    - Drill two 1/2” (1.27 cm) diameter holes 1” (2.54 cm) from the edge of the pipe
        - Oppose the 1/2” holes 180 deg. from each other
            - We aligned these to the center 2” holes above

Assembly of the Anchor Block

It is easier to install the eyebolts before we attach the 4” PVC pipe to the anchor block (**Figure 41**).

- Insert a 5/16” thread concrete anchor/lag shield into the hole
- With force, slowly twist in a 5/16” x 4” length stainless steel eye bolt into the lag shield

- Use the cold chisel (or a screwdriver) to continue to screw in the eye bolt approximately 2” (5.04 cm) deep into the lag shield (**Figure 41**).



Figure 41. Inserting the eye bolts into the lag screw shield in the anchor block.

Now we can install the pipe

- Center the 4” PVC Pipe on the block in the newly cut wide channel
- Using a 3/16” (0.47625 cm) concrete drill bit on a drill and through the 2” (5.08 cm) holes in the pipe:
  - Drill two pilot holes through the pipe and into the block to mark the locations of the pilot holes on the block
    - Remove the pipe
    - Continue to drill each these holes approximately 3” (7.62 cm) deep into the block
    - Replace the pipe on the block
      - The holes in the PVC pipe should align with the holes in the block. If they do not, you can easily re-drill new holes in the PVC as needed
- Place a 1/4” stainless steel fender washer over each of the anchor screws
- Using a drill with 5/16” driver:
  - Drive the screw and washer combo through the pipe holes and into the pilot holes in the concrete anchor block
  - DO NOT OVERDRIVE / OVERTIGHTEN THE ANCHOR SCREWS

Post-processing information

We provide two packages, *oceanwave*s and *owhlR*, that work with the statistical software R to help process raw CSV data files from the OWHL into wave statistics. Functions from the package *oce* are also used in the functions of the *oceanwaves* package, so *oce* must be installed as well.

The *oceanwaves* package can be found in the main R package repository at https://cran.r-project.org/package=oceanwaves, and *oce* can be found at https://cran.r-project.org/package=oce. Both packages can be installed from within an active R session using the commands:

```
install.packages('oceanwaves')
install.packages('oce')
```

The package *owhlR* can be downloaded as a .zip file from the permanent repository https://doi.org/10.5281/zenodo.3468467 or from https://github.com/millerlp/owhlR. The version found at Github can be installed from within an active R session using functions in the *devtools* package. The installation process would proceed as follows:

```
install.packages('devtools')
library(devtools)
install_github('millerlp/ohwlR', build_vignettes = TRUE)
```

If you want the latest development version of the *oceanwaves* package, you can instead install it as follows:

```
install_github('millerlp/oceanwaves', build_vignettes = TRUE)
```

The argument in the commands above to build vignettes will create the tutorial vignette for the *owhlR* package which details the current post-processing workflow. To load the *oceanwaves* and *owhlR* packages in a current R session, do the following:

```
library(oceanwaves)
library(owhlR)
```

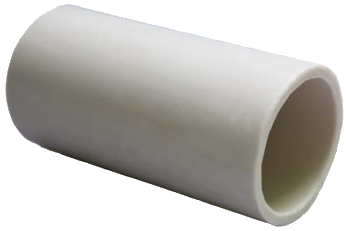
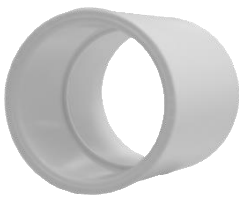
Once the *owhlR* package has been loaded, the vignette '`OWHL-workflow` can be loaded using the command:

```
vignette('OWHL-workflow', package = 'owhlR')
```

This will generate a html file in the user's browser that walks through the processing steps, with the relevant R code to generate wave statistics. If you get an error that the vignette can't be found, make sure you include the `build_vignettes=TRUE` argument when you installed *owhlR* from Github using the `install_github()` function above.

Materials lists

Table 4. PVC Housing Materials. PVC plumbing pipe and fittings are standardized, so while specific part numbers are given here, other brands or suppliers parts may be substituted. Quoted prices in US dollars are current for 2020.

Item 1	PVC DWV Plain End Pipe 1-1/2 in. x 10 ft. 330-PSI Schedule 40 Charlotte Pipe Model# 531111 $5.13 each (Model # and price from HomeDepot.com)	
Item 2	1-1/2 in. PVC Sch. 40 Socket Cap Charlotte Pipe Model# PVC021161400HD $1.14 each (Model # and price from HomeDepot.com)	
Item 3	1-1/2 in. PVC Slip x Slip Union Homewerks Worldwide Model# 511-14-112-112H $8.51 each (Model # and price from HomeDepot.com)	
Item 4	1-1/2 in. PVC DWV Hub x FIPT Female Adapter Charlotte Pipe Model# C4803HD112 $1.26 each (Model # and price from HomeDepot.com)	
Item 5	1-1/2 in. PVC DWV MIPT Cleanout Plug Charlotte Pipe Model # C4818HD112 $0.93 each (Model # and price from HomeDepot.com)	
Item 6	1-1/2 in. PVC Sch. 40 S x S Coupling Charlotte Pipe Model # PVC021001400HD $0.91 each (Model # and price from HomeDepot.com)	

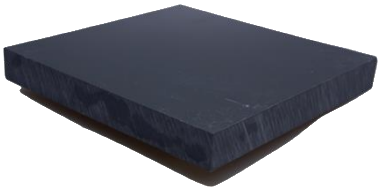
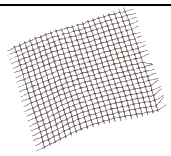
Item 7	1-1/2 in Clear PVC Pipe – Sch. 80 240 psi @ 72° F $25.55 (2') (McMaster-Carr # 4740K66)	
Item 8	PVC Sheet 12" x 12" x 3/4" $33.55 Each (McMaster-Carr #8747K117)	
Item 9	Copper Wire Cloth 16 x 16 Mesh Size, 0.052" Opening Size $8.34 (1ft x 1ft) (McMaster-Carr #9224T55)	
Item 10	8 oz. PVC Handy Pack Purple Primer and Solvent Cement Model # 302483 $9.17 (Model # and price from HomeDepot.com)	
Item 11	2 in. Bi-Metal Hole Saw Model # 49-56-9624 $10.97 (Model # and price from HomeDepot.com)	
Item 12	7/16 in. Large Hole Saw Arbor W/ Pilot Drill Bit $9-$20 (Amazon.com)	

Table 5. Oil Reservoir Materials. Quoted prices in US dollars are current for 2020.

Item 13	BAXTER IV Bags 0.9% Sodium Chloride Injection, USP, 25 mL VIAFLEX Plastic Container, Quad Pack (4 bags/overwrap). NDC 0338-0049-10 SKU: 2B1300 http://ecatalog.baxter.com (prices vary with distributor)	
Item 14	Plastic Barbed Tube Fittings for Air and Water Tight-Seal, Adapter, 3/16" Tube ID x 1/8 NPT Male, Black (McMaster-Carr #5463K11) $7.03 per pack of 10	
Item 15	Easy-Install Single Snap-Grip Clamp for Firm Hose and Tube, 1/4" to 5/16" ID (McMaster-Carr #5246K61) $5.90 per pack of 20	
Item 16	1/2 in. x 260 in. PTFE Tape Model# 0178502 98¢ each (Model # and price from HomeDepot.com)	
Item 17	8"(200mm) Impulse Manual Hand Bag Sealer Heat Sealing Machine for Plastics Bags ~$23-$28 (Amazon.com)	
Item 18	1/2" Industrial Liquid Dispenser Needle Adhesive Glue Dispensing Blunt Tip Luer Lock ~$7 (Amazon.com)	

Item 19	Squeeze bottle with large bore tip ~$6 (Amazon.com)	
Item 20	Plastic Syringe with Luer Lock Tip with Measurement, No Needle (10 ML) Multipack ~$12 (Amazon.com)	
Item 21	Mineral Oil (CVS.com, Walgreens.com) ~ $6	

Table 6. Miscellaneous Electronics Materials. Quoted prices in US dollars are current for 2020.

Item 22	Silica Desiccant Packets (Amazon.com) ~ $8 - $12 (bulk qty)	
Item 23	Battery Holder for 3 D Batteries Nickel-plated steel with solder lugs (McMaster-Carr #7712K55) $7.53 each	
Item 24	Header Pin, short (Amazon.com, eBay.com, Digikey.com) ~ $5 - $15 (bulk qty)	
Item 25	Header Pin, long (Amazon.com, eBay.com, Digikey.com) ~ $5 - $15 (bulk qty)	

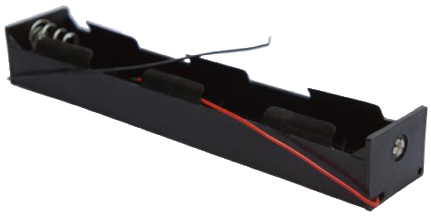
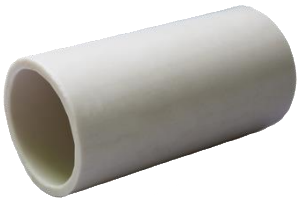
Item 26	CR 1220 3 volt Lithium button cell (Amazon.com, eBay.com, Digikey.com) ~ $4 - $8 (bulk qty)	
Item 27	Micro SD card (Amazon.com, eBay.com) ~ $5 - $55 (bulk qty)	
Item 28	Mini Micro JST 2.0 PH 2-Pin Connector Plug Male with 150mm Cable & Female (Amazon.com, eBay.com) ~ $3 - $9 (bulk qty)	
Item 29	*Battery Holder for 3 D Batteries* *Polypropylene End-to-End,* *Spring Holder* *(McMaster-Carr #7712K26* *$3.14 each* *(Replaced by Item 23 above)*	
Item 30	26 ga. Wire Spools (in red and black) (Amazon.com, eBay.com, Digikey.com) ~ $7 - $15	
Item 31	Heat Shrink Tubing (Amazon.com, eBay.com, Digikey.com) ~ $8 - $12 (bulk qty)	

Table 7. Miscellaneous OWHL Materials. Quoted prices in US dollars are current for 2020.

Item 31	3M 5200 Fast Cure Polyurethane Adhesive/Sealant Tube, White Model # 292592 | Mfg # 05220 UPC # 51135052204 $15.49 for 3 oz. (Westmarine.com)	
Item 32	Amazing GOOP Marine (Amazon.com, HomeDepot.com) ~ $5	
Item 33	Flanged Hex Head Screws for Sheet Metal Super-Corrosion-Resistant 316 Stainless Steel, Number 8 Size, 1/2" Long $9.09 per pack of 50 (McMaster-Carr #99663A213)	

Table 8. Materials for the Anchor Block. Quoted prices in US dollars are current for 2020.

Item 34	Concrete Deck Block Item # 560913 Model # 403216 (Common: 7-in x 11-in x 11-in; Actual: 7.125-in x 10.375-in x 10.375-in) ~$6 (Lowes.com)	
Item 35	PVC DWV Plain End Pipe 4 in. x 10 ft. 220-PSI Schedule 40 Charlotte Pipe Model# 531103 ~ $16.51 each (Model # and price from HomeDepot.com)	
Item 36	PVC DWV Hub Cap 4 in. x 10 ft. 220-PSI Schedule 40 Charlotte Pipe Model# C4817HD4 $2.34 each (Model # and price from HomeDepot.com)	
Item 37	5/16” x 2.1/4” Tapcon Concrete Screws (HomeDepot.com) ~$19.82 (box of 15)	
Item 38	5/16” x 4” Stainless Steel Eye Bolt (Lowes.com) ~ $2.50	
Item 39	1/4” Stainless Steel Fender Washers (Lowes.com) ~ $1.40 (x4)	

Item 40	3/16” Tapcon Concrete Drill bit (Lowes.com) ~$5.50	
Item 41	3/4” Concrete Drill bit (Lowes.com) ~ $8 to $40	
Item 42	Wood-Screw-Threaded Metal Anchors for 5/16" Screw Size, 1-3/4" Long (5/16” Lag Shield/ Concrete Anchor) $7.15 per pack of 10 (McMaster-Carr #97039A030)	

Wave sensor comparison table

Table 9. Comparison of specifications for the OWHL and commercial wave loggers.

	OWHL, MS5803-14BA pressure sensor	**Seabird SBE-26plus, strain gage pressure sensor**	**RBR Ltd. Virtuoso^3 D|wave 16**	**Datawell MkIII buoy, accelerometer sensors**
Parameter				
Pressure limit	1400 kPa (130 m depth; sensor only, housing not depth rated)	Varies by model, 300 to 6132 kPa (20 to 600 m depth)	300 or 603 kPa (20 or 50 m depth)	NA
Pressure, resolution	20 to 100 Pa (approx. 2-10 mm)	0.1 to 0.4 mm	<0.001% of full scale	NA
Pressure, accuracy	± 1.4% of full scale (± 20 mbar, ~ ± 200 mm)	± 0.1% of full scale (30 mm for 45psia model)	± 0.05% of full scale	NA
Pressure, stability	1.4% of full scale (20 mbar yr^{-1}, ~ 200 mm yr^{-1})	0.03% of full scale (9 mm for 45psia model)	0.05% of full scale	NA
Heave range	NA	NA	NA	± 20 m
Heave accuracy	NA	NA	NA	<0.5% of measured value
Sampling rate	1-4 Hz	1-4 Hz	1-16 Hz	3.84 Hz
Memory endurance	Flexible – 8GB microSD card yields ~ 500 d continuous 4 Hz sampling	32 MB 56.7 d sampling every 30 min for 1024 s 677 d sampling every 6 h for 1024 s	120M samples Up to 3477 d sampling continuously at 4 Hz	Flexible – 2 GB compact flash card yields 1095 d of spectral data and 3041 d of displacement data
Battery life	>14 months with 3 x D cell alkaline battery, continuous sampling at 4 Hz	Varies by sampling regime: 82 d with alkaline batteries, sampling every 30 min for 1024 s at 4 Hz 577 d with alkaline batteries, sampling every 6 h for 1024 s at 4 Hz	Battery life not quoted	14.5 months for 0.7 m diameter model, 43.5 months for 0.9 m diameter model, continuous sampling